ISC Class XII Mathematics

Chapter 5
Continuity and Differentiability

48 Types of Sums Solved

Step-by-Step ISC Board Solutions – ML Aggarwal

Sections Covered

I. Continuity – Direct Checking
II. Differentiability – Existence and Definition
III. Differentiation – Chain Rule and Standard Rules
IV. Differentiation of Inverse Trigonometric Functions
V. Implicit Differentiation
VI. Differentiation of Exponential and Logarithmic Functions
VII. Logarithmic Differentiation / Power-Tower Functions
VIII. Differentiation of Parametric Functions
IX. Second Order Derivatives

Compiled Notes – Formula Sheet + Type-wise Solved ISC Board Solutions

Contents

Index of Types

Type	Method / Pattern	★	Source(s)
Section I: Continuity – Direct Checking			
1	Checking continuity of a (often piecewise) function at a point	Yes	Ex 5.1 Q1(i); Ex 5.1 Q16(ii) (alt.)
2	Removable discontinuity: simplify a 0/0 rational form, classify & redefine	Yes	Ex 5.1 Q2; Ex 5.2 Q14 (alt.)
3	Finding one unknown constant k for continuity at a join	Yes	Ex 5.1 Q7; Ex 5.2 Q12 (ISC 2024) (alt.)
4	Finding a relationship / system between two constants a, b	Yes	Ex 5.1 Q21; Ex 5.1 Q22 (alt.)
5	Continuity using standard trig/log limit forms to find k	Yes	Ex 5.1 Q11; Ex 5.1 Q20(v) (alt.)
6	Combined join with different substitutions each side – find a AND b	–	Ex 5.1 Q23
7	Determining if ANY choice of $f(0)$ can give continuity	–	Ex 5.1 Q24

Type	Method / Pattern	★	Source(s)
8	Continuity via the Squeeze (Sandwich) Theorem	–	Ex 5.1 Q25
9	Continuity of a sum of modulus functions at corner points	–	Ex 5.1 Q26
10	One-sided discontinuity from exponential blow-up ($e^{1/x}$ type)	–	Ex 5.1 Q27
11	Continuity of the greatest integer function $[x]$	–	Ex 5.2 Q1
12	Locating ALL points of discontinuity over the domain	Yes	Ex 5.2 Q8(ii); Ex 5.2 Q9(i) (alt.)
13	Continuity via the algebra/composition-of-continuous-functions theorem	Yes	Ex 5.3 Q6(i); Ex 5.3 Q7(i) (alt.)
Section II: Differentiability – Existence and Definition			
14	Checking differentiability from the definition (LHD/RHD)	Yes	Ex 5.4 Q9; Ex 5.4 Q11 (alt.)
15	Proving continuous-but-not-differentiable at a modulus corner	Yes	Ex 5.4 Q13 (ISC 2019); Ex 5.4 Q14 (alt.)
16	Finding constants for BOTH continuity and differentiability at a join	Yes	Ex 5.4 Q18; Ex 5.4 Q21 (alt.)
17	Differentiability via a bounded-oscillation (squeeze-type) argument	Yes	Text Example 22; Ex 5.4 Q20(i) (alt.)
18	Using the derivative definition to evaluate a disguised limit	–	Ex 5.4 Q10
Section III: Differentiation – Chain Rule and Standard Rules			
19	Master chain rule for composite functions	Yes	Ex 5.5 Q7(i); Ex 5.5 Q3(ii) (NCERT) (alt.)
20	Differentiating modulus ($\|f(x)\|$) functions	Yes	Ex 5.5 Q10; Ex 5.5 Q11 (alt.)
21	Chain rule from a table of values (no explicit formula)	–	Ex 5.5 Q12 (ISC Specimen 2024)
22	Differentiate-then-prove a stated identity in dy/dx	Yes	Ex 5.5 Q19; Ex 5.5 Q20 (alt.)
23	Odd/even derivative proofs and functional-equation derivative tricks	Yes	Ex 5.5 Q22; Ex 5.5 Q23 (alt.)
24	Reading the derivative directly off a piecewise-linear graph	–	Ex 5.5 Q26
Section IV: Differentiation of Inverse Trigonometric Functions			
25	Domain of differentiability of the six inverse trig functions	–	Ex 5.6 Q1
26	Direct chain-rule differentiation of inverse trig compositions	Yes	Ex 5.6 Q2(iv); Ex 5.6 Q10(i) (alt.)
27	Simplifying via inverse-trig identities before differentiating	Yes	Ex 5.6 Q5; Ex 5.6 Q4 (alt.)
28	Simplifying via trigonometric substitution before differentiating	Yes	Ex 5.6 Q16(i); Ex 5.6 Q18(ii) (NCERT) (alt.)

Type	Method / Pattern	★	Source(s)
Section V: Implicit Differentiation			
29	Standard implicit differentiation of an equation in x, y	Yes	Ex 5.7 Q8(v); Ex 5.7 Q8(vi) (alt.)
30	Verifying $\frac{dy}{dx} \cdot \frac{dx}{dy} = 1$ via implicit differentiation	–	Ex 5.7 Q7(i)
31	Differentiating an infinite nested radical (self-referential equation)	Yes	Ex 5.7 Q12(i); Ex 5.7 Q12(ii) (alt.)
Section VI: Differentiation of Exponential and Logarithmic Functions			
32	Direct chain rule for exponential/log composite functions	Yes	Ex 5.8 Q4(i) (NCERT); Ex 5.8 Q6(ii) (alt.)
33	Checking derivability of a log function at a boundary point	–	Ex 5.8 Q7
34	Differentiating an abstract function using given $f(a), f'(a)$	Yes	Ex 5.8 Q10; Ex 5.8 Q11 (alt.)
35	Simplifying an exponential expression via log rules before differentiating	–	Ex 5.8 Q18(ii)
36	Implicit differentiation mixing exponential and log terms	Yes	Ex 5.8 Q20(ii); Ex 5.8 Q21 (alt.)
37	Differentiating a log with a variable base (change of base first)	Yes	Ex 5.8 Q23(i); Ex 5.8 Q23(ii) (alt.)
Section VII: Logarithmic Differentiation / Power-Tower Functions			
38	Logarithmic differentiation of a product of several factors	Yes	Ex 5.9 Q1(i) (NCERT); Ex 5.9 Q1(ii) (alt.)
39	Master power-tower type $[f(x)]^{g(x)}$	Yes	Ex 5.9 Q3(ii); Ex 5.9 Q4(ii) (alt.)
40	Distinguishing x^x, a^x, x^a, a^a in one combined sum	Yes	Ex 5.9 Q7 (NCERT); Ex 5.9 Q8(iii) (alt.)
41	Implicit power-tower equations (e.g. $x^y = y^x$)	Yes	Ex 5.9 Q10(ii); Ex 5.9 Q10(vi) (alt.)
42	Infinite power-tower (self-referential exponent)	–	Ex 5.9 Q11
Section VIII: Differentiation of Parametric Functions			
43	Standard parametric differentiation $dy/dx = (dy/dt)/(dx/dt)$	Yes	Ex 5.10 Q1(ii) (NCERT); Ex 5.10 Q7(i) (alt.)
44	Differentiating one function w.r.t. a different (non-x) function	Yes	Ex 5.10 Q12(ii); Ex 5.10 Q12(iv) (alt.)
45	Differentiating one inverse-trig expression w.r.t. another	Yes	Ex 5.10 Q16(ii); Ex 5.10 Q13(i) (alt.)
Section IX: Second Order Derivatives			
46	Direct second derivative of an explicit function	Yes	Ex 5.11 Q1(iii) (NCERT); Ex 5.11 Q4(iv) (NCERT) (alt.)
47	Proving a stated second-order differential-equation-style identity	Yes	Ex 5.11 Q12(i); Ex 5.11 Q12(ii) (ISC 2024) (alt.)
48	Second derivative of parametric functions	Yes	Ex 5.11 Q24 (ISC Specimen 2024); Ex 5.11 Q21 (alt.)

Formula Sheet

Section I – Continuity

$$f \text{ is continuous at } x = c \iff \lim_{x\to c^-} f(x) = \lim_{x\to c^+} f(x) = f(c)$$

LHL = RHL = value of the function at the point

Section I – Standard Limits

$$\lim_{x\to 0}\frac{\sin x}{x} = 1,\ \lim_{x\to 0}\frac{\tan x}{x} = 1,\ \lim_{x\to 0}\frac{\log(1+x)}{x} = 1,\ \lim_{x\to 0}\frac{e^x-1}{x} = 1$$

The four standard limit forms used to evaluate continuity limits

Section I – Squeeze Theorem

$$\text{If } g(x) \le f(x) \le h(x) \text{ near } c \text{ and } \lim g(x) = \lim h(x) = L, \text{ then } \lim f(x) = L$$

Used when a function is bounded-oscillating × vanishing (e.g. $x\sin(1/x)$)

Section II – Differentiability

$$f'_-(c) = \lim_{h\to 0^-}\frac{f(c+h)-f(c)}{h}, \quad f'_+(c) = \lim_{h\to 0^+}\frac{f(c+h)-f(c)}{h}$$

Left-hand and right-hand derivatives; f is differentiable at c iff both exist and are equal

Section II – Key Theorem

$$\text{Differentiable at } c \implies \text{continuous at } c \quad \text{(converse is FALSE)}$$

The single most important conceptual fact in this chapter

Section III – Chain Rule

$$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$$

Differentiate the outer function, keep the inner function unchanged, then multiply by the derivative of the inner function

Section III – Modulus Rule

$$\frac{d}{dx}|u| = \frac{u}{|u|} \cdot \frac{du}{dx}, \quad u \neq 0$$

Equivalent to $\mathrm{sgn}(u) \cdot u'$

Section IV – Inverse Trig Derivatives

$$\frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}\cos^{-1}x = \frac{-1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}\tan^{-1}x = \frac{1}{1+x^2}$$

The three most commonly needed inverse trig derivatives

Section IV – Inverse Trig Derivatives

$$\frac{d}{dx}\cot^{-1}x = \frac{-1}{1+x^2}, \quad \frac{d}{dx}\sec^{-1}x = \frac{1}{|x|\sqrt{x^2-1}}, \quad \frac{d}{dx}\csc^{-1}x = \frac{-1}{|x|\sqrt{x^2-1}}$$

The remaining three inverse trig derivatives

Section IV – Common Substitutions

$$x = \tan\theta \ (\text{for } 1+x^2); \quad x = \sin\theta \ (\text{for } 1-x^2); \quad x = \cos 2\theta \ (\text{for } 1 \pm x)$$

Substitutions that simplify inverse-trig expressions before differentiating

Section V – Implicit Differentiation

$$\text{Differentiate both sides w.r.t. } x, \text{ treating } y \text{ as a function of } x, \text{ then solve}$$

Apply the chain rule to every term containing y, remembering $\frac{d}{dx}(y^n) = ny^{n-1}\frac{dy}{dx}$

Section VI – Exponential/Log Derivatives

$$\frac{d}{dx}e^x = e^x, \quad \frac{d}{dx}\log x = \frac{1}{x}, \quad \frac{d}{dx}a^x = a^x \log a$$

The three base exponential/log derivative rules

Section VI – Change of Base

$$\log_a b = \frac{\log b}{\log a}$$

Used to convert a variable-base logarithm into natural logs before differentiating

Section VII – Logarithmic Differentiation

$$\text{For } y = [f(x)]^{g(x)} : \text{ take } \log \text{ of both sides} \Rightarrow \log y = g(x) \log f(x), \text{ then differentiate implicitly}$$

Essential whenever the variable appears in BOTH the base and the exponent

Section VIII – Parametric Differentiation

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \quad \frac{dx}{dt} \neq 0$$

Differentiate x and y separately with respect to the parameter t, then divide

Section IX – Second Order Derivative

$$y_2 = \frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$

Differentiate the first derivative again with respect to x

Section IX – Second Order Parametric

$$\frac{d^2y}{dx^2} = \frac{d}{dt}\left(\frac{dy}{dx}\right) \div \frac{dx}{dt}$$

Never differentiate dy/dx directly w.r.t. x for parametric functions – always go back through t

I. Continuity – Direct Checking

Type 1

Checking whether a (often piecewise) function is continuous at a given point by computing the left-hand limit (LHL), right-hand limit (RHL), and the function's actual value at that point, then comparing all three.

★ **Board Favorite** – This is the foundational continuity-checking skill, tested in nearly every ISC paper – the book tags this exact question format directly to NCERT.

Formula Used: f continuous at $x = c \iff \lim_{x\to c^-} f(x) = \lim_{x\to c^+} f(x) = f(c)$

Source: Exercise 5.1, Q1(i) (NCERT)

Given: $f(x) = \begin{cases} x^3 + 3, & x \neq 0 \\ 1, & x = 0 \end{cases}$ at $x = 0$

Solution:

Step 1: Compute the left-hand limit: $\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-}(x^3 + 3) = 0 + 3 = 3$ – using the rule for $x \neq 0$, since points just left of 0 are still $\neq 0$.

Step 2: Compute the right-hand limit: $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+}(x^3 + 3) = 3$ – same rule applies from the right too.

Step 3: Note the actual function value: $f(0) = 1$ – read directly from the second piece of the definition.

Step 4: Compare: LHL $= 3 =$ RHL, but $f(0) = 1 \neq 3$ – the limit exists but doesn't match the function's actual value there.

Final Answer

f is discontinuous at $x = 0$ (the limit exists and equals 3, but $f(0) = 1 \neq 3$)

Common Mistake: Stopping after showing LHL=RHL and concluding "continuous" – matching one-sided limits is necessary but NOT sufficient; the common limit must also equal $f(c)$ itself.

Alternate Board-Style Example

Source: Exercise 5.1, Q16(ii)

Given: $f(x) = \begin{cases} 5x - 4, & x < 1 \\ 4x^2 - 3x, & x \geq 1 \end{cases}$ at $x = 1$

Solution:

Step 1: LHL: $\lim_{x\to 1^-}(5x - 4) = 5(1) - 4 = 1$.

Step 2: RHL: $\lim_{x\to 1^+}(4x^2 - 3x) = 4(1) - 3(1) = 1$.

Step 3: $f(1)$: since $1 \geq 1$, use the second piece: $f(1) = 4(1)^2 - 3(1) = 1$.

Step 4: All three values – LHL, RHL, and $f(1)$ – equal 1.

Final Answer

f is continuous at $x = 1$

Quick Recall: Always compute all three quantities separately (LHL, RHL, $f(c)$) and explicitly compare all three – never assume the function value just because the limits agree.

Type 2

A function is given as a 0/0-form rational expression at one point (with a separately-assigned value there) – simplify the rational expression by cancelling the common factor to test continuity, and if discontinuous, classify the type of discontinuity and redefine the function to remove it.

★ **Board Favorite –** Removable-discontinuity questions (simplify, classify, redefine) are a recurring board pattern – the book tags a fuller version of this exact task as a Competency Focused Question.

Source: Exercise 5.1, Q2

Given: $f(x) = \dfrac{x^2 - 1}{x - 1}$ for $x \neq 1$, $\quad f(1) = 2$. Show continuous at $x = 1$.

Solution:

Step 1: Factor the numerator: $x^2 - 1 = (x - 1)(x + 1)$ – standard difference of squares.

Step 2: Cancel the common factor (valid since $x \neq 1$ throughout the limit process): $\dfrac{(x-1)(x+1)}{x-1} = x + 1$.

Step 3: So $\lim_{x\to 1} f(x) = \lim_{x\to 1}(x + 1) = 2$ – this single simplified limit serves as both LHL and RHL since the simplified expression is a plain polynomial (continuous everywhere).

Step 4: Compare with $f(1) = 2$ (given) – they match.

Final Answer

f is continuous at $x = 1$, since $\lim_{x \to 1} f(x) = 2 = f(1)$

Common Mistake: Cancelling $(x - 1)$ and treating the simplified function $x + 1$ as identical to the original f – they agree everywhere EXCEPT possibly at $x = 1$ itself, which is exactly the point being tested, so the assigned value $f(1)$ must still be checked separately.

Alternate Board-Style Example (fuller, multi-part)

Source: Exercise 5.2, Q14

Given: $f(x) = \dfrac{x^2 - 9}{x - 3}$, $x \neq 3$, $\quad f(3) = 5$

Solution:

Case: (i) Is there a removable discontinuity at $x = 3$?

Factor and cancel: $\dfrac{x^2 - 9}{x - 3} = \dfrac{(x - 3)(x + 3)}{x - 3} = x + 3$ for $x \neq 3$, so $\lim_{x \to 3} f(x) = 3 + 3 = 6$. Since $f(3) = 5 \neq 6$, the limit exists but doesn't match the assigned value – yes, there is a removable discontinuity.

Case: (ii) Classify it

Because the limit exists (both one-sided limits agree and are finite) but simply doesn't match the given function value, this is a *missing point* (or *isolated point*) discontinuity – the graph has a single punctured/misplaced point, not a jump or an infinite blow-up.

Case: (iii) Redefine to make it continuous

Simply reassign the value at the single problem point to match the limit: redefine $f(3) = 6$ instead of 5.

Final Answer

(i) Yes, removable discontinuity at $x = 3$ (ii) Missing/isolated point discontinuity (iii) Redefine $f($
6

Quick Recall: Whenever you see a rational function with a common factor, factor and cancel first – the resulting simplified limit tells you what value the function "should" have; compare that to what's actually assigned to spot and fix a removable discontinuity.

Type 3

Finding a single unknown constant k so that a piecewise function becomes continuous at the join point – set the limit from one side equal to the value/expression on the other side and solve for k.

★ **Board Favorite** – "Find k for continuity" is one of the most repeated VSA formats in ISC papers – the book tags a version of this directly to ISC 2024.

Source: Exercise 5.1, Q7

Given: $f(x) = \begin{cases} kx^2 + 5, & x \leq 1 \\ 2, & x > 1 \end{cases}$, find k so that f is continuous at $x = 1$

Solution:

Step 1: For continuity at $x = 1$, the limit from the left must equal the limit from the right (both then automatically equal $f(1)$, since $f(1)$ is given by the left piece here).

Step 2: LHL: $\lim_{x\to 1^-}(kx^2 + 5) = k(1)^2 + 5 = k + 5$.

Step 3: RHL: $\lim_{x\to 1^+} 2 = 2$.

Step 4: Set LHL = RHL: $k + 5 = 2 \Rightarrow k = -3$.

Final Answer

$k = -3$

Common Mistake: Plugging $x = 1$ into the WRONG piece, or forgetting that since $x \leq 1$ uses the first piece, $f(1)$ itself is automatically $k + 5$ (no separate check needed once LHL=RHL here).

Alternate Board-Style Example

Source: Exercise 5.2, Q12 (ISC 2024)

Given: $f(x) = \begin{cases} \dfrac{(x+3)^2 - 36}{x - 3}, & x \neq 3 \\ k, & x = 3 \end{cases}$, continuous at $x = 3$

Solution:

Step 1: Expand the numerator: $(x + 3)^2 - 36 = x^2 + 6x + 9 - 36 = x^2 + 6x - 27$.

Step 2: Factor: $x^2 + 6x - 27 = (x - 3)(x + 9)$ – checking: $(x - 3)(x + 9) = x^2 + 9x - 3x - 27 = x^2 + 6x - 27$. Correct.

Step 3: Cancel: $\dfrac{(x-3)(x+9)}{x-3} = x+9$ for $x \neq 3$.

Step 4: $\lim_{x\to 3} f(x) = 3+9 = 12$. For continuity, k must equal this limit.

Final Answer

$k = 12$

Quick Recall: Set the two one-sided expressions (or the simplified limit vs. the assigned constant) equal to each other and solve – this is just Type 1's comparison, run in reverse to find the unknown instead of verifying a known one.

Type 4

Finding a relationship between TWO unknown constants (a and b) for continuity at a join – either a single equation relating them (if only one join point is given), or a full system of two equations (if the join must hold at an interior point sandwiched between two pieces, giving two separate matching conditions).

★ **Board Favorite –** Multi-constant continuity problems are a recurring long-answer board favorite, testing the same LHL=RHL=f(c) idea but requiring simultaneous-equation solving.

Source: Exercise 5.1, Q21

Given: $f(x) = \begin{cases} ax+1, & x \leq 3 \\ bx+3, & x > 3 \end{cases}$, continuous at $x = 3$

Solution:

Step 1: LHL: $\lim_{x\to 3^-}(ax+1) = 3a+1$.

Step 2: RHL: $\lim_{x\to 3^+}(bx+3) = 3b+3$.

Step 3: Set LHL = RHL for continuity: $3a+1 = 3b+3$.

Step 4: Simplify: $3a = 3b+2$ – this is as far as we can go with only one condition; the relationship, not unique values, is what's asked.

Final Answer

$3a = 3b + 2$

Common Mistake: Trying to solve for unique numeric values of a and b when the question only gives ONE join point (hence only one equation) – with a single condition and two unknowns, only a relationship between them can be found, not specific values.

Alternate Board-Style Example (system of two equations)

Source: Exercise 5.1, Q22

Given: $f(x) = \begin{cases} 3ax + b, & x > 1 \\ 11, & x = 1 \\ 5ax - 2b, & x < 1 \end{cases}$, continuous at $x = 1$. Find a, b.

Solution:

Step 1: Here there IS a specific value $f(1) = 11$ given, so we get TWO separate equations: LHL$= f(1)$ and RHL$= f(1)$.

Step 2: LHL: $\lim_{x\to1^-}(5ax - 2b) = 5a - 2b$. Set equal to 11: $5a - 2b = 11$.

Step 3: RHL: $\lim_{x\to1^+}(3ax + b) = 3a + b$. Set equal to 11: $3a + b = 11$.

Step 4: Now solve the system: from the second equation, $b = 11 - 3a$. Substitute into the first: $5a - 2(11 - 3a) = 11 \Rightarrow 5a - 22 + 6a = 11 \Rightarrow 11a = 33 \Rightarrow a = 3$.

Step 5: Back-substitute: $b = 11 - 3(3) = 11 - 9 = 2$.

Final Answer

$a = 3,\ b = 2$

Quick Recall: Count how many "matching conditions" the join actually gives you – one join with no separate assigned value gives ONE equation (a relationship only); a join with a specific middle value given gives TWO equations (solvable for unique a, b).

Type 5

Finding an unknown constant k using the STANDARD trigonometric or logarithmic limit forms ($\frac{\sin\theta}{\theta} \to 1$, $\frac{\tan\theta}{\theta} \to 1$, etc.) rather than plain algebraic substitution – requires massaging the expression into the exact standard form first.

★ **Board Favorite** – Using the standard trig-limit toolkit to solve for k is one of the highest-frequency VSA patterns in the whole continuity topic.

Formula Used: $\lim_{\theta\to0} \dfrac{\tan\theta}{\theta} = 1, \quad \lim_{\theta\to0} \dfrac{\sin\theta}{\theta} = 1$

Source: Exercise 5.1, Q11

Given: $f(x) = \begin{cases} \dfrac{\tan 3x}{kx}, & x \neq 0 \\ 1, & x = 0 \end{cases}$, find k so that f is continuous at $x = 0$

Solution:

Step 1: For continuity, $\lim_{x\to 0} f(x) = f(0) = 1$.

Step 2: Rewrite $\dfrac{\tan 3x}{kx}$ to expose the standard form: multiply and divide by 3 inside, $\dfrac{\tan 3x}{kx} = \dfrac{3}{k} \cdot \dfrac{\tan 3x}{3x}$ – engineering the argument of tan to match the denominator.

Step 3: As $x \to 0$, also $3x \to 0$, so $\dfrac{\tan 3x}{3x} \to 1$ by the standard limit.

Step 4: So $\lim_{x\to 0} f(x) = \dfrac{3}{k} \cdot 1 = \dfrac{3}{k}$. Set this equal to $f(0) = 1$: $\dfrac{3}{k} = 1 \Rightarrow k = 3$.

Final Answer

$k = 3$

Common Mistake: Forgetting to multiply and divide to match the argument inside tan/sin with the denominator – $\frac{\tan(3x)}{x}$ is NOT directly 1; only $\frac{\tan(3x)}{3x} \to 1$.

Alternate Board-Style Example

Source: Exercise 5.1, Q20(v)

Given: $f(x) = \begin{cases} \dfrac{\tan 5x}{\sin 2x}, & x \neq 0 \\ k, & x = 0 \end{cases}$, continuous at $x = 0$

Solution:

Step 1: Rewrite using both standard forms at once: $\dfrac{\tan 5x}{\sin 2x} = \dfrac{\tan 5x}{5x} \cdot \dfrac{2x}{\sin 2x} \cdot \dfrac{5x}{2x}$ – engineering both the tan and sin arguments to match their own denominators, with a leftover algebraic ratio.

Step 2: As $x \to 0$: $\dfrac{\tan 5x}{5x} \to 1$ and $\dfrac{2x}{\sin 2x} \to 1$ (reciprocal of the standard sine limit).

Step 3: The leftover ratio simplifies: $\dfrac{5x}{2x} = \dfrac{5}{2}$.

Step 4: So $\lim_{x\to 0} f(x) = 1 \cdot 1 \cdot \dfrac{5}{2} = \dfrac{5}{2}$, which must equal k.

Final Answer

$k = \dfrac{5}{2}$

Quick Recall: Whenever you see $\frac{\sin(\cdot)}{(\cdot)}$ or $\frac{\tan(\cdot)}{(\cdot)}$ with mismatched arguments, multiply and divide by whatever constant makes the argument inside the trig function match its own denominator exactly – then the standard limit $\to 1$ applies cleanly.

Type 6

A join point where the LEFT piece and RIGHT piece each need a different limit-technique (e.g. a trig-limit expansion on one side, a rationalisation on the other), and BOTH an unknown a (inside the left piece) and b (inside the right piece) must be found together from the single continuity condition plus the given middle value.

Source: Exercise 5.1, Q23

Given: $f(x) = \begin{cases} \dfrac{\sin((a+1)x) + 2\sin x}{x}, & x < 0 \\ 2, & x = 0 \\ \dfrac{\sqrt{1+bx}-1}{x}, & x > 0 \end{cases}$, continuous at $x = 0$. Find a, b.

Solution:

Step 1: **Left-hand limit:** split the numerator into two standard-limit pieces: $\dfrac{\sin((a+1)x)}{x} + \dfrac{2\sin x}{x}$.

Step 2: For the first piece, multiply and divide by $(a+1)$: $\dfrac{\sin((a+1)x)}{x} = (a+1) \cdot \dfrac{\sin((a+1)x)}{(a+1)x} \to (a+1) \cdot 1 = a+1$ as $x \to 0$.

Step 3: The second piece: $\dfrac{2\sin x}{x} \to 2 \cdot 1 = 2$ directly.

Step 4: So LHL $= (a+1) + 2 = a + 3$. Set this equal to $f(0) = 2$: $a + 3 = 2 \Rightarrow a = -1$.

Step 5: **Right-hand limit:** rationalise $\dfrac{\sqrt{1+bx}-1}{x}$ by multiplying by the conjugate $\dfrac{\sqrt{1+bx}+1}{\sqrt{1+bx}+1}$: $\dfrac{(1+bx)-1}{x(\sqrt{1+bx}+1)} = \dfrac{bx}{x(\sqrt{1+bx}+1)} = \dfrac{b}{\sqrt{1+bx}+1}$.

Step 6: As $x \to 0$: $\dfrac{b}{\sqrt{1+0}+1} = \dfrac{b}{2}$. Set this equal to $f(0) = 2$: $\dfrac{b}{2} = 2 \Rightarrow b = 4$.

Final Answer

$a = -1,\ b = 4$

Common Mistake: Trying to apply the SAME technique (e.g. only trig-limit expansion) to both sides – always identify which specific technique (standard trig limit vs. rationalisation) each side's algebraic FORM actually calls for.

Quick Recall: Treat the two sides of the join completely independently – each gives its own equation in its own unknown by matching to the common middle value $f(0)$; don't expect the same trick to work on both sides.

Type 7

Determining whether ANY choice of $f(0)$ (or the value at the problem point) can make a 0/0-type function continuous there – sometimes a valid choice exists (LHL=RHL, just assign that common value), and sometimes NO choice works because the left and right limits are fundamentally different.

Source: Exercise 5.1, Q24

Given: (i) $f(x) = \dfrac{\sqrt[3]{1+x}-1}{x}$ (ii) $f(x) = \dfrac{5|x| - 3\tan x + \sin 2x}{x}$, $x \neq 0$ in both cases. What choice of $f(0)$ (if any) gives continuity?

Solution:

Case: (i) $f(x) = \dfrac{\sqrt[3]{1+x}-1}{x}$

Use the standard limit $\lim_{x\to 0} \dfrac{(1+x)^n - 1}{x} = n$ with $n = \frac{1}{3}$ (a generalised version of the binomial-limit form): $\lim_{x\to 0} \dfrac{(1+x)^{1/3}-1}{x} = \dfrac{1}{3}$. Since this limit exists (and is the same from both sides, as the expression is identical on both sides here), assigning $f(0) = \dfrac{1}{3}$ makes f continuous.

Case: (ii) $f(x) = \dfrac{5|x| - 3\tan x + \sin 2x}{x}$

Because of the $|x|$ term, the LEFT and RIGHT limits must be checked separately. For $x > 0$: $|x| = x$, so $f(x) = \dfrac{5x - 3\tan x + \sin 2x}{x} = 5 - 3 \cdot \dfrac{\tan x}{x} + 2 \cdot \dfrac{\sin 2x}{2x} \to 5 - 3(1) + 2(1) = 4$ as $x \to 0^+$. For $x < 0$: $|x| = -x$, so $f(x) = \dfrac{-5x - 3\tan x + \sin 2x}{x} = -5 - 3 \cdot \dfrac{\tan x}{x} + 2 \cdot \dfrac{\sin 2x}{2x} \to -5 - 3(1) + 2(1) = -6$ as $x \to 0^-$. Since $4 \neq -6$, the left and right limits genuinely disagree.

Final Answer

$(i) f(0) = \dfrac{1}{3}$ works (ii) NO choice of $f(0)$ can make it continuous, since LHL $= -6 \neq 4 =$ RHL

Common Mistake: Treating $|x|$ as simply "x" without splitting into the left ($|x| = -x$) and right ($|x| = x$) cases – this is the exact source of the mismatch in part (ii), and missing it leads to wrongly concluding a value of $f(0)$ exists.

Quick Recall: Whenever $|x|$ (or any modulus) appears with $x \to 0$, ALWAYS split into the $x > 0$ and $x < 0$ cases before taking the limit – if the two one-sided limits disagree, no assigned value can ever fix the discontinuity.

Type 8

Proving continuity at a point using the Squeeze (Sandwich) Theorem – useful when the function is a bounded oscillating factor (like $\sin(1/x)$ or $\cos(1/x)$) multiplied by a factor that vanishes at the point in question.

Formula Used: If $g(x) \leq f(x) \leq h(x)$ near c and $\lim g(x) = \lim h(x) = L$, then $\lim f(x) = L$

Source: Exercise 5.1, Q25 (Competency Focused Question)

Given: $f(x) = \begin{cases} x\sin\dfrac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, examine continuity at $x = 0$

Solution:

Step 1: The function $\sin(1/x)$ oscillates rapidly near $x = 0$ and has no limit there by itself, but it is always bounded: $-1 \leq \sin\dfrac{1}{x} \leq 1$ for every $x \neq 0$ – this bound is the key to applying the squeeze theorem.

Step 2: Multiply the inequality by $|x|$ (which is ≥ 0, so the inequality direction is preserved when we take absolute values throughout): $-|x| \leq x\sin\dfrac{1}{x} \leq |x|$.

Step 3: As $x \to 0$, both bounding functions $-|x|$ and $|x|$ tend to 0 – so by the squeeze theorem, the function sandwiched between them must also tend to 0.

Step 4: So $\lim_{x\to 0} f(x) = 0$, and this matches $f(0) = 0$ exactly as given.

Final Answer

f is continuous at $x = 0$

Common Mistake: Trying to evaluate $\lim_{x\to 0}\sin(1/x)$ directly and treating it as some specific number – this limit does NOT exist by itself; the squeeze theorem is needed precisely because the oscillating part alone has no limit, but gets pinned to 0 once multiplied by a vanishing factor.

Quick Recall: Whenever you see (bounded oscillating trig of $1/x$) $\times$ (something $\to 0$), reach for the squeeze theorem: bound the oscillating part between -1 and 1, multiply through by the vanishing factor, and let both outer bounds collapse to the same limit.

Type 9

Checking continuity of a function built from a SUM of modulus (absolute value) terms at the "corner" points where the individual modulus expressions change their sign/definition.

Source: Exercise 5.1, Q26 (Competency Focused Question)

Given: $f(x) = |x| + |x-1|,\ x \in \mathbf{R}$. Show continuous at both $x = 0$ and $x = 1$.

Solution:

Step 1: Break the function into pieces based on where each modulus changes sign: for $x < 0$, $|x| = -x$ and $|x-1| = -(x-1) = 1-x$, so $f(x) = -x+1-x = 1-2x$. For $0 \leq x < 1$, $|x| = x$ and $|x-1| = 1-x$, so $f(x) = x+1-x = 1$. For $x \geq 1$, $|x| = x$ and $|x-1| = x-1$, so $f(x) = x+x-1 = 2x-1$.

Step 2: **At $x = 0$:** LHL $= \lim_{x\to 0^-}(1-2x) = 1$; RHL $= \lim_{x\to 0^+} 1 = 1$; $f(0) = |0|+|0-1| = 0+1 = 1$ – all three match.

Step 3: **At $x = 1$:** LHL $= \lim_{x\to 1^-} 1 = 1$; RHL $= \lim_{x\to 1^+}(2x-1) = 2(1)-1 = 1$; $f(1) = |1|+|1-1| = 1+0 = 1$ – all three match again.

Final Answer

f is continuous at both $x = 0$ and $x = 1$

Common Mistake: Assuming a function built from modulus terms must automatically be discontinuous at the "corner" points – corners can break DIFFERENTIABILITY (see Section II) without breaking continuity; the two properties must be checked separately.

Quick Recall: Split a sum of modulus terms into its piecewise-linear form across each critical point first, then run the ordinary LHL/RHL/f(c) check at every corner – absolute value sums are usually continuous everywhere even where they're not smooth.

Type 10

Examining continuity of a function involving $e^{1/x}$ (or similar one-sided exponential blow-up) at the point where the exponent becomes undefined – the key insight is that $1/x \to +\infty$ from one side and $1/x \to -\infty$ from the other, sending $e^{1/x}$ to two completely different limits.

Source: Exercise 5.1, Q27

Given: $f(x) = \begin{cases} e^{1/x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$, examine continuity at $x = 0$

Solution:

Step 1: As $x \to 0^-$ (from the left, negative side), $\dfrac{1}{x} \to -\infty$ – since dividing 1 by a very small negative number gives a very large negative number.

Step 2: So $e^{1/x} \to e^{-\infty} = 0$ – an exponential with an exponent going to $-\infty$ shrinks to 0. This gives LHL $= 0$.

Step 3: As $x \to 0^+$ (from the right, positive side), $\dfrac{1}{x} \to +\infty$, so $e^{1/x} \to e^{+\infty} = +\infty$ – the right-hand limit doesn't even exist as a finite number.

Step 4: Since LHL $(= 0)$ and RHL (does not exist / is infinite) don't even agree in kind, let alone value, f cannot be continuous at $x = 0$ – and this conclusion has nothing to do with what $f(0)$ was assigned to be.

Final Answer

f is discontinuous at $x = 0$, regardless of the value assigned to $f(0)$, since the right-hand limit doesn'
∞)

Common Mistake: Assuming that since ONE side (the left) gives a nice finite limit, the function might still be "fixable" by choosing the right $f(0)$ – once one side's limit fails to exist at all, no assignment of $f(0)$ can ever create continuity.

Quick Recall: Whenever $1/x$ sits inside an exponential near $x = 0$, always check the LEFT ($1/x \to -\infty$, so $e^{1/x} \to 0$) and RIGHT ($1/x \to +\infty$, so $e^{1/x} \to \infty$) sides separately – they almost never agree.

Type 11

Checking continuity of the greatest integer (floor) function $f(x) = [x]$ at a given point – integer points are where jumps occur, while non-integer points are always safely inside a "flat" piece.

Source: Exercise 5.2, Q1

Given: $f(x) = [x]$. Check continuity at (i) $x = 0$ (ii) $x = 1$ (iii) $x = \dfrac{1}{2}$

Solution:

Case: (i) At $x = 0$ (an integer point)

For x slightly less than 0 (e.g. $x = -0.1$), $[x] = -1$, so LHL $= -1$. For x slightly more than 0 (e.g. $x = 0.1$), $[x] = 0$, so RHL $= 0$. Since LHL $\neq$ RHL, f is discontinuous at $x = 0$.

Case: (ii) At $x = 1$ (an integer point)
By the exact same reasoning, LHL $= 0$ (just below 1) but RHL $= 1$ (at and just above 1) – discontinuous again.

Case: (iii) At $x = \frac{1}{2}$ (a non-integer point)
For x near $\frac{1}{2}$ (both slightly less and slightly more, as long as we stay between 0 and 1), $[x] = 0$ throughout – so LHL $=$ RHL $= f(\frac{1}{2}) = 0$, all matching.

Final Answer

(i) Discontinuous at $x = 0$ (ii) Discontinuous at $x = 1$ (iii) Continuous at $x = \frac{1}{2}$

Common Mistake: Assuming the greatest integer function is discontinuous "everywhere" just because it's a step function – it is actually continuous at every NON-integer point, and only jumps at integers.

Quick Recall: $[x]$ jumps by exactly 1 at every integer (discontinuous there) and is perfectly flat (hence continuous) at every non-integer point in between.

Type 12

Locating ALL the points of discontinuity of a function across its entire domain, rather than testing continuity at just one pre-specified point – requires identifying every "suspicious" point (piecewise joins, removable factors) and checking each one.

★ **Board Favorite –** "Locate the points of discontinuity" (rather than checking a single given point) is a recurring exercise format that tests whether a student can independently identify WHERE to check, not just how.

Source: Exercise 5.2, Q8(ii) (NCERT)

Given: $f(x) = \begin{cases} x^3 - 3, & x \leq 2 \\ x^2 + 1, & x > 2 \end{cases}$

Solution:

Step 1: Both pieces ($x^3 - 3$ and $x^2 + 1$) are polynomials, which are continuous everywhere on their own – so the ONLY point that could possibly cause trouble is the join point, $x = 2$.

Step 2: LHL: $\lim_{x\to 2^-}(x^3 - 3) = 8 - 3 = 5$.

Step 3: RHL: $\lim_{x\to 2^+}(x^2 + 1) = 4 + 1 = 5$.

Step 4: $f(2)$: since $2 \leq 2$, use the first piece: $f(2) = 8 - 3 = 5$.

Step 5: All three agree, so $x = 2$ is actually fine.

Final Answer

f has NO points of discontinuity – it is continuous everywhere

Common Mistake: Assuming a piecewise function defined by two different formulas MUST have a discontinuity somewhere – it's entirely possible (as here) for the pieces to join up perfectly smoothly.

Alternate Board-Style Example (a removable point this time)

Source: Exercise 5.2, Q9(i)

Given: $f(x) = \begin{cases} \dfrac{x^4 - 16}{x - 2}, & x \neq 2 \\ 16, & x = 2 \end{cases}$

Solution:

Step 1: Away from $x = 2$, this is a quotient of polynomials with a nonzero denominator, hence continuous everywhere except possibly at $x = 2$ – so again, only ONE point needs checking.

Step 2: Factor: $x^4 - 16 = (x^2 - 4)(x^2 + 4) = (x - 2)(x + 2)(x^2 + 4)$.

Step 3: Cancel: $\dfrac{(x-2)(x+2)(x^2+4)}{x-2} = (x + 2)(x^2 + 4)$ for $x \neq 2$.

Step 4: $\lim_{x\to 2} f(x) = (2 + 2)(4 + 4) = 4 \times 8 = 32$.

Step 5: Compare with the assigned value $f(2) = 16$ – they do NOT match ($32 \neq 16$).

Final Answer

f has exactly one point of discontinuity, at $x = 2$ (a removable/missing-point discontinuity, since the

Quick Recall: For a piecewise function built from otherwise-continuous pieces (polynomials, simplifiable rationals), the only candidates for discontinuity are the join points and any points where a denominator could vanish – check just those, not the whole domain point-by-point.

Type 13

Determining continuity of a function by recognising it as a sum, difference, product, quotient, or COMPOSITION of functions that are already known to be continuous – using the algebra-of-continuous-functions theorem directly, without computing any limits by hand.

★ **Board Favorite** – This "reason about continuity using the theorem" format (rather than grinding through limits) is a common quick VSA/objective pattern, and also appears as a Competency Focused proof-style question.

Formula Used: Sums, differences, products, quotients (denominator $\neq 0$), and compositions of continuous functions are continuous

Source: Exercise 5.3, Q6(i) (Competency Focused Question)

Given: Prove that $\dfrac{2x^3 - 7x^2 + 3}{(x-1)(x+3)}$ is continuous.

Solution:

Step 1: The numerator $2x^3 - 7x^2 + 3$ is a polynomial, and polynomials are continuous everywhere on $\mathbf{R}$ – a standard known fact, requiring no separate proof.

Step 2: The denominator $(x-1)(x+3)$ is also a polynomial, hence continuous everywhere too.

Step 3: By the quotient rule for continuous functions, a quotient of two continuous functions is continuous everywhere EXCEPT where the denominator is zero.

Step 4: The denominator is zero exactly when $x = 1$ or $x = -3$ – these are excluded from the domain of f in the first place, so within its actual domain, f is continuous throughout.

Final Answer

The function is continuous at every point of its domain (i.e. every real number except $x = 1$ and $x = -3$)

Common Mistake: Trying to compute LHL/RHL by hand at every single point of the domain – when the function is built purely from known-continuous pieces via sum/product/quotient, the algebra-of-continuous-functions theorem lets you skip direct limit computation entirely.

Alternate Board-Style Example (composition)

Source: Exercise 5.3, Q7(i) (NCERT)

Given: Examine $\cos(x^2)$ for continuity.

Solution:

Step 1: Recognise this as a COMPOSITION: the inner function is $g(x) = x^2$ and the outer function is $f(u) = \cos u$.

Step 2: Both $g(x) = x^2$ (a polynomial) and $f(u) = \cos u$ (a standard trig function) are known to be continuous everywhere on their respective domains.

Step 3: By the theorem "a composition of continuous functions is continuous," $f(g(x)) = \cos(x^2)$ is therefore continuous everywhere.

Final Answer

$\cos(x^2)$ is continuous for all $x \in \mathbf{R}$

Quick Recall: Before reaching for limits, first ask: "can I build this function out of pieces I already KNOW are continuous, using $+, -, \times, \div$, or composition?" If yes, the algebra-of-continuous-functions theorem finishes the proof instantly.

II. Differentiability – Existence and Definition

Type 14

Checking differentiability of a piecewise function at the join point directly from the definition – compute the left-hand derivative (LHD) and right-hand derivative (RHD) using the limit definition, and compare.

★ **Board Favorite –** This is the foundational differentiability-checking skill – nearly every long-answer question in this topic ultimately reduces to computing LHD and RHD from the definition.

Formula Used: $f'_-(c) = \lim_{h\to 0^-} \dfrac{f(c+h)-f(c)}{h}$, $f'_+(c) = \lim_{h\to 0^+} \dfrac{f(c+h)-f(c)}{h}$

Source: Exercise 5.4, Q9 (Exemplar)

Given: $f(x) = \begin{cases} 1+x, & x \le 2 \\ 5-x, & x > 2 \end{cases}$, examine differentiability at $x = 2$

Solution:

Step 1: First confirm $f(2)$ from the definition: since $2 \le 2$, $f(2) = 1 + 2 = 3$.

Step 2: **LHD:** for small negative h, $2+h \le 2$, so use the first piece: $f'_-(2) = \lim_{h\to 0^-} \dfrac{f(2+h)-f(2)}{h} = \lim_{h\to 0^-} \dfrac{(1+2+h)-3}{h} = \lim_{h\to 0^-} \dfrac{h}{h} = 1$.

Step 3: **RHD:** for small positive h, $2+h > 2$, so use the second piece: $f'_+(2) = \lim_{h\to 0^+} \dfrac{f(2+h)-f(2)}{h} = \lim_{h\to 0^+} \dfrac{(5-(2+h))-3}{h} = \lim_{h\to 0^+} \dfrac{-h}{h} = -1$.

Step 4: Since $f'_-(2) = 1 \ne -1 = f'_+(2)$, the two one-sided derivatives disagree.

Final Answer

f is NOT differentiable at $x = 2$ (LHD = 1, RHD = -1, which disagree)

Common Mistake: Using the same piece of the function for both the LHD and RHD computation – always re-check which piece is "active" for small negative h versus small positive h around the join point.

Alternate Board-Style Example (continuity AND differentiability, two different points)

Source: Exercise 5.4, Q11

Given: $f(x) = \begin{cases} 5x-4, & 0<x<1 \\ 4x^2-3x, & 1 \le x < 2 \\ 3x+4, & x \ge 2 \end{cases}$, examine continuity at $x = 1$ and differentiability at $x = 2$

Solution:

Case: Continuity at $x = 1$

LHL: $\lim_{x\to 1^-}(5x-4) = 1$. RHL: $\lim_{x\to 1^+}(4x^2-3x) = 4-3 = 1$. $f(1) = 4(1)-3(1) = 1$. All three agree – continuous at $x = 1$.

Case: Differentiability at $x = 2$

$f(2) = 3(2)+4 = 10$ (using the third piece, active at $x \ge 2$). LHD: $f'_-(2) = \lim_{h\to 0^-}\dfrac{f(2+h)-f(2)}{h} = \lim_{h\to 0^-}\dfrac{[4(2+h)^2-3(2+h)]-10}{h}$; expanding $4(2+h)^2-3(2+h) = 4(4+4h+h^2)-6-3h = 16+16h+4h^2-6-3h = 10+13h+4h^2$, so the numerator is $13h+4h^2$, giving $f'_-(2) = \lim_{h\to 0^-}(13+4h) = 13$. RHD: $f'_+(2) = \lim_{h\to 0^+}\dfrac{[3(2+h)+4]-10}{h} = \lim_{h\to 0^+}\dfrac{3h}{h} = 3$.

Final Answer

f is continuous at $x = 1$; f is NOT differentiable at $x = 2$ (LHD = 13, RHD = 3)

Quick Recall: Always identify which piece of the definition is "active" just to the left and just to the right of the point in question before substituting into the LHD/RHD limit formulas – a wrong piece choice silently ruins the whole computation.

Type 15

Proving that a modulus-based function (like $|x-a|$, or a sum of such terms) is continuous at a "corner" point but specifically NOT differentiable there – the graph has a sharp kink, so the two one-sided slopes are finite but different.

★ **Board Favorite –** The classic $|x-a|$ continuous-but-not-differentiable proof is one of the most-tested conceptual results in the chapter, with the book tagging a direct instance to ISC 2019.

Source: Exercise 5.4, Q13 (ISC 2019)

Given: $f(x) = |x-4|$. Show continuous but not differentiable at $x = 4$.

Solution:

Step 1: **Continuity:** $\lim_{x\to 4^-} |x-4| = \lim_{x\to 4^-}(4-x) = 0$ (since $x < 4$ makes $x-4 < 0$, so $|x-4| = 4-x$); $\lim_{x\to 4^+} |x-4| = \lim_{x\to 4^+}(x-4) = 0$ (since $x > 4$ makes $|x-4| = x-4$); and $f(4) = |4-4| = 0$. All three agree – continuous at $x = 4$.

Step 2: **LHD:** $f'_-(4) = \lim_{h\to 0^-} \dfrac{f(4+h)-f(4)}{h} = \lim_{h\to 0^-} \dfrac{|h|-0}{h}$. Since $h < 0$ here, $|h| = -h$, giving $\lim_{h\to 0^-} \dfrac{-h}{h} = -1$.

Step 3: **RHD:** $f'_+(4) = \lim_{h\to 0^+} \dfrac{|h|-0}{h}$. Since $h > 0$ here, $|h| = h$, giving $\lim_{h\to 0^+} \dfrac{h}{h} = 1$.

Step 4: LHD $= -1 \neq 1 =$ RHD – the one-sided derivatives disagree.

Final Answer

f is continuous at $x = 4$ but NOT differentiable there (LHD = -1, RHD = 1)

Common Mistake: Assuming that because the function IS continuous, it should also be differentiable – continuity is a NECESSARY but not sufficient condition for differentiability; a sharp corner is the classic counter-example.

Alternate Board-Style Example (two corners)

Source: Exercise 5.4, Q14

Given: $f(x) = |x-3| + |x-4|$. Show not differentiable at $x = 3$ and $x = 4$.

Solution:

Case: At $x = 3$

$f'_-(3)$: for $h < 0$ small, both $3+h-3 = h < 0$ and $3+h-4 < 0$, so $f(3+h) = -(h) + (1-h) = -h + (4-3-h)$... more directly, near $x = 3$ (with $x < 4$ always), $|x-4| = 4-x$ is smooth (differentiable, slope -1), so the kink at $x = 3$ comes entirely from $|x-3|$: its LHD is -1 and RHD is $+1$ (same pattern as the main example). So $f'_-(3) = -1 + (-1) = -2$ and $f'_+(3) = 1 + (-1) = 0$ – combining the kink from $|x-3|$ with the smooth slope -1 of $|x-4| = 4-x$ at that point.

Case: At $x = 4$

Similarly, near $x = 4$ (with $x > 3$ always there), $|x - 3| = x - 3$ is smooth with slope $+1$, so the kink comes from $|x - 4|$: $f'_-(4) = 1 + (-1) = 0$ and $f'_+(4) = 1 + 1 = 2$.

Final Answer

f is not differentiable at $x = 3$ (LHD=-2, RHD=0) or at $x = 4$ (LHD=0, RHD=2)

Quick Recall: Each $|x - a|$ term contributes a slope of -1 just left of a and $+1$ just right of a; add up the (smooth, unchanging) contributions from the OTHER terms separately to get the full one-sided derivatives at each corner.

Type 16

Finding unknown constants so that a piecewise function is differentiable across its whole domain – this always requires TWO conditions at the join: matching function VALUES (for continuity, a prerequisite) AND matching DERIVATIVES (for differentiability itself).

★ **Board Favorite –** This combined continuity-and-differentiability constants problem is one of the highest-value long-answer question types in the whole chapter.

Source: Exercise 5.4, Q18

Given: $f(x) = \begin{cases} ax + b, & 0 < x \leq 1 \\ 2x^2 - x, & 1 < x < 2 \end{cases}$ is differentiable in $(0, 2)$. Find a, b.

Solution:

Step 1: **Continuity condition at $x = 1$:** the two pieces must give the same value there: $a(1) + b = 2(1)^2 - 1 \Rightarrow a + b = 1$.

Step 2: **Differentiability condition at $x = 1$:** differentiate each piece and match at $x = 1$. The derivative of $ax + b$ is simply a (a constant slope). The derivative of $2x^2 - x$ is $4x - 1$, which at $x = 1$ gives $4(1) - 1 = 3$.

Step 3: Set the two derivatives equal: $a = 3$.

Step 4: Substitute back into the continuity equation: $3 + b = 1 \Rightarrow b = -2$.

Final Answer

$a = 3,\ b = -2$

Common Mistake: Solving only the continuity equation and forgetting the SECOND condition (matching derivatives) entirely – one equation with two unknowns can never pin down unique values; differentiability always contributes the second equation needed.

Alternate Board-Style Example

Source: Exercise 5.4, Q21 (Competency Focused Question)

Given: $f(x) = \begin{cases} ax+b, & x > -1 \\ bx^2-3, & x \le -1 \end{cases}$ is differentiable for all x. Find a, b.

Solution:

Step 1: **Continuity at $x = -1$:** $a(-1)+b = b(-1)^2 - 3 \Rightarrow -a+b = b-3 \Rightarrow -a = -3 \Rightarrow a = 3$.

Step 2: **Differentiability at $x = -1$:** derivative of $ax+b$ is a; derivative of bx^2-3 is $2bx$, which at $x = -1$ gives $-2b$.

Step 3: Set the derivatives equal: $a = -2b$. Since $a = 3$ from Step 1, this gives $3 = -2b \Rightarrow b = -\frac{3}{2}$.

Final Answer

$a = 3,\ b = -\frac{3}{2}$

Quick Recall: Always write down BOTH equations before solving anything – (1) plug the join point into both pieces and set them equal for continuity, (2) differentiate both pieces separately and set THOSE equal at the join point for differentiability – then solve the resulting system together.

Type 17

Examining both continuity and differentiability at a point using a bounded-oscillation argument (like $x\cos(1/x)$ or $x\sin(1/x)$) – continuity often holds via the squeeze theorem, but differentiability frequently fails because the derivative computation reintroduces the un-squeezable oscillating term directly.

★ **Board Favorite –** This "continuity holds via squeeze, but differentiability fails" pattern is a favorite because it tests the crucial distinction between the two properties in one single question.

Source: Text Example 22

Given: $f(x) = \begin{cases} x\cos\frac{1}{x}, & x > 0 \\ 0, & x \le 0 \end{cases}$, examine continuity and differentiability at $x = 0$

Solution:

Step 1: **Continuity:** for $x > 0$, $\left|\cos\dfrac{1}{x}\right| \leq 1$, so by the squeeze theorem (as in Type 8), $\lim_{x\to 0^+} x\cos\dfrac{1}{x} = 0$. Also $\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} 0 = 0$. Both match $f(0) = 0$ – continuous.

Step 2: **LHD:** $f'_-(0) = \lim_{h\to 0^-} \dfrac{f(h) - f(0)}{h} = \lim_{h\to 0^-} \dfrac{0-0}{h} = 0$ – since $f(h) = 0$ for $h \leq 0$ by definition.

Step 3: **RHD:** $f'_+(0) = \lim_{h\to 0^+} \dfrac{f(h) - f(0)}{h} = \lim_{h\to 0^+} \dfrac{h\cos(1/h) - 0}{h} = \lim_{h\to 0^+} \cos\dfrac{1}{h}$ – the h cancels, but this leaves the RAW oscillating $\cos(1/h)$ with NOTHING left to squeeze it to a single value.

Step 4: Since $\cos(1/h)$ oscillates between -1 and 1 infinitely often as $h \to 0^+$ without settling on any single value, this limit does not exist.

Final Answer

f is continuous at $x = 0$, but NOT differentiable at $x = 0$ (RHD does not exist)

Common Mistake: Assuming that because the squeeze theorem worked for continuity, the same "bounded, so it's fine" reasoning applies to the derivative – differentiating cancels the vanishing factor that made the squeeze possible in the first place, leaving a bare oscillating term with no limit.

Alternate Board-Style Example (exercise version)

Source: Exercise 5.4, Q20(i)

Given: $f(x) = \begin{cases} x\sin\dfrac{1}{x}, & x < 0 \\ 0, & x \geq 0 \end{cases}$, examine continuity and differentiability at $x = 0$

Solution:

Step 1: **Continuity:** by the squeeze theorem exactly as before, $\lim_{x\to 0^-} x\sin(1/x) = 0$, matching $f(0) = 0$ and the trivial right-hand limit – continuous at $x = 0$.

Step 2: **LHD:** $f'_-(0) = \lim_{h\to 0^-} \dfrac{h\sin(1/h) - 0}{h} = \lim_{h\to 0^-} \sin\dfrac{1}{h}$, which oscillates without settling – does not exist.

Step 3: **RHD:** $f'_+(0) = \lim_{h\to 0^+} \dfrac{0-0}{h} = 0$.

Final Answer

f is continuous at $x = 0$, but NOT differentiable there (LHD does not exist)

Quick Recall: "Bounded oscillation $\times$ vanishing factor" gives continuity via the squeeze theorem, but differentiating removes the vanishing factor and exposes the raw oscillation – so differentiability almost always fails at exactly that point.

Type 18

Using the definition of the derivative directly to evaluate a given limit expression that isn't written in the standard $\frac{f(c+h)-f(c)}{h}$ form – recognising the disguised structure first is the key step.

Source: Exercise 5.4, Q10

Given: $f(2) = 4,\ f'(2) = 4$. Evaluate $\lim_{x\to 2} \dfrac{xf(2) - 2f(x)}{x-2}$

Solution:

Step 1: The expression doesn't look like the derivative definition yet – add and subtract a helper term to reveal the hidden structure: $xf(2) - 2f(x) = xf(2) - 2f(2) + 2f(2) - 2f(x) = f(2)(x-2) + 2(f(2) - f(x))$ – splitting the numerator into a piece proportional to $(x-2)$ and a piece resembling $-(f(x) - f(2))$.

Step 2: Divide by $(x-2)$: $\dfrac{xf(2) - 2f(x)}{x-2} = f(2) - 2 \cdot \dfrac{f(x) - f(2)}{x-2}$ – now the second term is EXACTLY the derivative-definition form (with $x \to 2$ playing the role of $c+h \to c$).

Step 3: Take the limit as $x \to 2$: the first term is just the constant $f(2) = 4$. The second term's fraction tends to $f'(2) = 4$ by definition.

Step 4: So the limit becomes $4 - 2(4) = 4 - 8 = -4$.

Final Answer

-4

Common Mistake: Trying to substitute $x = 2$ directly into the original expression without first splitting it – this gives the indeterminate form $\frac{0}{0}$ since both $xf(2) - 2f(x)$ and $x - 2$ vanish at $x = 2$; the algebraic split is what reveals the derivative in disguise.

Quick Recall: Whenever a limit as $x \to c$ produces $\frac{0}{0}$ and involves $f(x) - f(c)$ (or can be engineered to), add/subtract a term to isolate $\dfrac{f(x) - f(c)}{x-c}$ exactly – that piece is just $f'(c)$ by definition.

III. Differentiation – Chain Rule and Standard Rules

Type 19

Applying the chain rule to differentiate a composite function, possibly with several layers of "function inside a function" – differentiate from the outside in, one layer at a time.

★ **Board Favorite** – The chain rule is the single most-used tool in this entire chapter – every other differentiation technique (implicit, logarithmic, parametric) is built on top of it.

Formula Used: $\dfrac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$

Source: Exercise 5.5, Q7(i)

Given: $y = \cos(\tan\sqrt{x+1})$

Solution:

Step 1: Identify the layers from outside in: $\cos(\cdot)$, then $\tan(\cdot)$, then $\sqrt{\cdot}$, then $x+1$ – four nested layers to peel back one at a time.

Step 2: Differentiate the outermost cosine layer, keeping its argument unchanged: $\dfrac{dy}{dx} = -\sin(\tan\sqrt{x+1}) \cdot \dfrac{d}{dx}\left(\tan\sqrt{x+1}\right)$.

Step 3: Differentiate the tan layer: $\dfrac{d}{dx}\tan\sqrt{x+1} = \sec^2\sqrt{x+1} \cdot \dfrac{d}{dx}\sqrt{x+1}$.

Step 4: Differentiate the square-root layer: $\dfrac{d}{dx}\sqrt{x+1} = \dfrac{1}{2\sqrt{x+1}} \cdot \dfrac{d}{dx}(x+1) = \dfrac{1}{2\sqrt{x+1}} \cdot 1 = \dfrac{1}{2\sqrt{x+1}}$.

Step 5: Multiply all the layers together: $\dfrac{dy}{dx} = -\sin(\tan\sqrt{x+1}) \cdot \sec^2\sqrt{x+1} \cdot \dfrac{1}{2\sqrt{x+1}}$.

Final Answer

$$\frac{dy}{dx} = \frac{-\sin(\tan\sqrt{x+1})\sec^2\sqrt{x+1}}{2\sqrt{x+1}}$$

Common Mistake: Stopping after differentiating only the outermost layer and forgetting to keep multiplying by the derivative of each subsequent inner layer – with n nested layers, there must be n multiplied factors in the final answer.

Alternate Board-Style Example

Source: Exercise 5.5, Q3(ii) (NCERT)

Given: $y = \sin^3 x + \cos^6 x$

Solution:

Step 1: Differentiate term by term. For $\sin^3 x = (\sin x)^3$: treat this as (outer) cube of (inner) $\sin x$: $\dfrac{d}{dx}(\sin x)^3 = 3(\sin x)^2 \cdot \cos x = 3\sin^2 x \cos x$.

Step 2: For $\cos^6 x = (\cos x)^6$: similarly, $\dfrac{d}{dx}(\cos x)^6 = 6(\cos x)^5 \cdot (-\sin x) = -6\cos^5 x \sin x$.

Step 3: Add the two derivatives together: $\dfrac{dy}{dx} = 3\sin^2 x \cos x - 6\cos^5 x \sin x$.

Step 4: Factor out the common terms $3\sin x \cos x$: $= 3\sin x \cos x(\sin x - 2\cos^4 x)$.

Final Answer

$\dfrac{dy}{dx} = 3\sin x \cos x(\sin x - 2\cos^4 x)$

Quick Recall: Peel the composite function apart one layer at a time from the OUTSIDE in, differentiating each layer while temporarily freezing everything inside it – then multiply all the individual layer-derivatives together.

Type 20

Differentiating a function that involves a modulus (absolute value) expression, using $\frac{d}{dx}|u| = \frac{u}{|u|} \cdot u'$ (equivalently, the sign of u times u').

★ Board Favorite – Modulus differentiation tests whether a student remembers that $|u|$ is NOT simply differentiated like u – a distinct sign-based rule applies, making it a recurring board favorite.

Formula Used: $\dfrac{d}{dx}|u| = \dfrac{u}{|u|} \cdot \dfrac{du}{dx},\ u \neq 0$

Source: Exercise 5.5, Q10

Given: Find the derivative of $|2x^2 - 3|$ w.r.t. x.

Solution:

Step 1: Let $u = 2x^2 - 3$, so the function is $|u|$.

Step 2: Apply the modulus derivative rule: $\dfrac{d}{dx}|u| = \dfrac{u}{|u|} \cdot \dfrac{du}{dx}$.

Step 3: Compute $\dfrac{du}{dx} = 4x$.

Step 4: Substitute back: $\dfrac{d}{dx}|2x^2 - 3| = \dfrac{2x^2 - 3}{|2x^2 - 3|} \cdot 4x = \dfrac{4x(2x^2 - 3)}{|2x^2 - 3|}$.

Final Answer

$$\frac{4x(2x^2 - 3)}{|2x^2 - 3|}$$

Common Mistake: Simply writing the derivative as $4x$ (i.e. differentiating $2x^2 - 3$ and ignoring the modulus entirely) – the modulus introduces a sign factor $\frac{u}{|u|}$ that must always be carried through.

Alternate Board-Style Example

Source: Exercise 5.5, Q11 (Competency Focused Question)

Given: $f(x) = |\cos x|$, find $f'\left(\dfrac{3\pi}{4}\right)$

Solution:

Step 1: Using the modulus rule with $u = \cos x$: $f'(x) = \dfrac{\cos x}{|\cos x|} \cdot (-\sin x) = \dfrac{-\sin x \cos x}{|\cos x|}$.

Step 2: Evaluate the sign of $\cos x$ at $x = \dfrac{3\pi}{4}$: since $\dfrac{3\pi}{4}$ is in the second quadrant, $\cos \dfrac{3\pi}{4} = -\dfrac{1}{\sqrt{2}} < 0$, so $|\cos x| = -\cos x$ there.

Step 3: Substitute: $f'\left(\dfrac{3\pi}{4}\right) = \dfrac{-\sin(3\pi/4)\cos(3\pi/4)}{-\cos(3\pi/4)} = \sin\left(\dfrac{3\pi}{4}\right)$ – the cos terms cancel (with sign), leaving just sin.

Step 4: $\sin \dfrac{3\pi}{4} = \dfrac{1}{\sqrt{2}}$.

Final Answer

$f'\left(\frac{3\pi}{4}\right) = \frac{1}{\sqrt{2}}$

Quick Recall: Never differentiate straight through a modulus sign – always introduce the $\frac{u}{|u|}$ sign factor first, then substitute the specific point (checking which sign u actually has there) only at the very end.

Type 21

Using the chain rule when u and v (and their derivatives) are given only as a TABLE of numeric values at specific points – no explicit formula exists, so the chain rule must be applied purely by matching up the correct table entries.

Source: Exercise 5.5, Q12 (ISC Specimen Paper 2024)

Given:

x	$u(x)$	$u'(x)$	$v(x)$	$v'(x)$
1	0	2	-1	5
2	4	5	3	2
3	2	-1	3	0

$r(x) = u(v(x))$, find $r'(2)$

Solution:

Step 1: By the chain rule, $r'(x) = u'(v(x)) \cdot v'(x)$ – this is the general formula, before plugging in any specific numbers.

Step 2: We need $r'(2)$, so substitute $x = 2$: $r'(2) = u'(v(2)) \cdot v'(2)$.

Step 3: Read from the table at $x = 2$: $v(2) = 3$ and $v'(2) = 2$.

Step 4: Since $v(2) = 3$, we need $u'(3)$ (NOT $u'(2)$) – this is the step most easily missed: the chain rule needs u' evaluated AT $v(2)$, not at $x = 2$ itself. Reading the table at $x = 3$: $u'(3) = -1$.

Step 5: Multiply: $r'(2) = u'(3) \cdot v'(2) = (-1)(2) = -2$.

Final Answer

$r'(2) = -2$

Common Mistake: Reading $u'(2)$ straight from the row $x = 2$ instead of first finding $v(2)$ and THEN looking up u' at that value – this is the classic table-based chain-rule trap.

Quick Recall: For $r(x) = u(v(x))$, first look up $v(2)$ in the table, then use THAT value (not 2 itself) to look up u'; separately look up $v'(2)$ directly; multiply the two.

Type 22

Differentiating a given relation between x and y and then algebraically simplifying to PROVE that $\frac{dy}{dx}$ satisfies some stated identity – the differentiation is usually the easy part; the algebra afterward is where the identity gets confirmed.

★ **Board Favorite –** "Differentiate, then prove the identity" is one of the most common long-answer formats in this chapter, since it tests both mechanical differentiation AND algebraic simplification skill together.

Source: Exercise 5.5, Q19

Given: $y = \sqrt{x} + \dfrac{1}{\sqrt{x}}$. Show that $2x\dfrac{dy}{dx} + y = 2\sqrt{x}$

Solution:

Step 1: Rewrite using fractional powers to make differentiation cleaner: $y = x^{1/2} + x^{-1/2}$.

Step 2: Differentiate term by term: $\dfrac{dy}{dx} = \dfrac{1}{2}x^{-1/2} - \dfrac{1}{2}x^{-3/2}$.

Step 3: Multiply both sides by $2x$ (as the identity requires): $2x\dfrac{dy}{dx} = 2x\left(\dfrac{1}{2}x^{-1/2}\right) - 2x\left(\dfrac{1}{2}x^{-3/2}\right) = x^{1/2} - x^{-1/2}$ – simplifying the exponents ($x \cdot x^{-1/2} = x^{1/2}$, and $x \cdot x^{-3/2} = x^{-1/2}$).

Step 4: Add $y = x^{1/2}+x^{-1/2}$ to both sides: $2x\dfrac{dy}{dx}+y = (x^{1/2}-x^{-1/2})+(x^{1/2}+x^{-1/2}) = 2x^{1/2}$ – the $x^{-1/2}$ terms cancel exactly.

Step 5: Rewrite $2x^{1/2}$ back as $2\sqrt{x}$.

Final Answer

$2x\dfrac{dy}{dx} + y = 2\sqrt{x},$ as required

Common Mistake: Trying to verify the identity by plugging in one specific numeric value of x instead of a full algebraic proof – "show that" questions need the identity demonstrated for ALL valid x, not just checked for one number.

Alternate Board-Style Example

Source: Exercise 5.5, Q20

Given: $y = \sqrt{\dfrac{1-\sin 2x}{1+\sin 2x}}$. Prove that $\dfrac{dy}{dx} + \sec^2\left(\dfrac{\pi}{4} - x\right) = 0$

Solution:

Step 1: Simplify the expression under the root FIRST using $1 \pm \sin 2x = (\cos x \pm \sin x)^2$ (a standard trig identity, since $(\cos x \pm \sin x)^2 = 1 \pm 2\sin x\cos x = 1 \pm \sin 2x$): so $y = \sqrt{\dfrac{(\cos x - \sin x)^2}{(\cos x + \sin x)^2}} = \left|\dfrac{\cos x - \sin x}{\cos x + \sin x}\right|$, which (for the relevant domain) simplifies to $y = \dfrac{\cos x - \sin x}{\cos x + \sin x}$.

Step 2: Divide numerator and denominator by $\cos x$: $y = \dfrac{1-\tan x}{1+\tan x}$ – this is exactly the tangent subtraction formula in disguise: $y = \tan\left(\dfrac{\pi}{4} - x\right)$.

Step 3: Differentiate this much simpler form: $\dfrac{dy}{dx} = \sec^2\left(\dfrac{\pi}{4} - x\right) \cdot \dfrac{d}{dx}\left(\dfrac{\pi}{4} - x\right) = \sec^2\left(\dfrac{\pi}{4} - x\right) \cdot (-1) = -\sec^2\left(\dfrac{\pi}{4} - x\right)$.

Step 4: Add $\sec^2\left(\dfrac{\pi}{4} - x\right)$ to both sides: $\dfrac{dy}{dx} + \sec^2\left(\dfrac{\pi}{4} - x\right) = 0$.

Final Answer

$\dfrac{dy}{dx} + \sec^2\left(\dfrac{\pi}{4} - x\right) = 0$, as required

Quick Recall: Before differentiating a messy trig expression, always check whether it secretly simplifies via a standard identity (like $1 \pm \sin 2x = (\cos x \pm \sin x)^2$, or the tan subtraction formula) – simplify FIRST, differentiate SECOND.

Type 23

Proving general derivative facts about odd/even functions, or using a clever substitution trick ($x \to -x$) on a functional equation involving $f(x)$ and $f(-x)$ to solve for a specific derivative value.

★ **Board Favorite –** The functional-equation "differentiate, then substitute x and $-x$ to get simultaneous equations" trick is a distinctive, recurring higher-order board question.

Source: Exercise 5.5, Q22

Given: Prove that the derivative of an odd function is always an even function.

Solution:

Step 1: Let f be odd, meaning by definition $f(-x) = -f(x)$ for all x – this is the given general property to work from.

Step 2: Differentiate both sides of $f(-x) = -f(x)$ with respect to x, applying the chain rule to the left side (since $-x$ is a function of x): $\frac{d}{dx}f(-x) = f'(-x)\cdot(-1) = -f'(-x)$.

Step 3: The right side differentiates directly: $\frac{d}{dx}(-f(x)) = -f'(x)$.

Step 4: Equate the two sides: $-f'(-x) = -f'(x) \Rightarrow f'(-x) = f'(x)$.

Step 5: By definition, a function g satisfying $g(-x) = g(x)$ is even – and that's exactly what we've shown for f'.

Final Answer

$f'(-x) = f'(x),$ so the derivative of an odd function is even

Common Mistake: Forgetting to apply the chain rule (the extra factor of -1) when differentiating $f(-x)$ with respect to x – this sign is exactly what flips the parity from odd to even.

Alternate Board-Style Example (functional equation trick)

Source: Exercise 5.5, Q23

Given: $2f(x) + 3f(-x) = x^2 + x + 1$. Find $f'(1)$.

Solution:

Step 1: Differentiate the given equation with respect to x, remembering the chain rule for $f(-x)$: $2f'(x) + 3f'(-x)\cdot(-1) = 2x + 1$, i.e. $2f'(x) - 3f'(-x) = 2x + 1$. (Equation A)

Step 2: This single equation still has two unknowns ($f'(x)$ and $f'(-x)$), so substitute $x = 1$ into Equation A: $2f'(1) - 3f'(-1) = 3$. (Equation B)

Step 3: To get a second independent equation, substitute $x = -1$ into Equation A instead: $2f'(-1) - 3f'(1) = -1$. (Equation C)

Step 4: Now solve Equations B and C simultaneously. From B: $f'(-1) = \frac{2f'(1) - 3}{3}$. Substitute into C: $2\left(\frac{2f'(1) - 3}{3}\right) - 3f'(1) = -1 \Rightarrow \frac{4f'(1) - 6}{3} - 3f'(1) = -1$.

Step 5: Multiply through by 3: $4f'(1) - 6 - 9f'(1) = -3 \Rightarrow -5f'(1) = 3 \Rightarrow f'(1) = -\frac{3}{5}$.

Final Answer

$f'(1) = -\dfrac{3}{5}$

Common Mistake: Substituting only ONE value of x (e.g. just $x = 1$) and trying to solve for $f'(1)$ alone – since the differentiated equation always has both $f'(x)$ and $f'(-x)$ together, TWO substitutions ($x = 1$ and $x = -1$) are needed to form a solvable system.

Quick Recall: For odd/even proofs, differentiate the defining equation and watch the chain-rule sign flip parity; for "$af(x)+bf(-x) = \ldots$" style equations, differentiate once, then substitute both $x = c$ and $x = -c$ to get a solvable pair of simultaneous equations.

Type 24

Reading the derivative of a function directly off a piecewise-linear graph (e.g. a V-shape or zig-zag) without ever being given an explicit formula – the derivative is simply the SLOPE of each linear segment, valid on the open interval between corners.

Source: Exercise 5.5, Q26

Given: The graph of $y = f(x)$ is a V-shape with its vertex at $(-3, 0)$, rising to height 3 on the y-ax

Solution:

Step 1: Since the graph is made of straight-line segments, the derivative on each segment is simply that segment's constant slope – no differentiation formulas are needed, just reading the picture.

Step 2: For $x < -3$ (the left arm of the V, descending as we move right toward the vertex, or equivalently rising steeply as read left-to-right from the vertex outward): the segment falls from a positive height down to 0 at $x = -3$ as x increases toward -3 from further left... reading the standard V-graph convention, the left arm has a NEGATIVE slope of -1 (since it comes down from upper-left to the vertex) – matching the marked right angle indicating the two arms are perpendicular, each at 45 to the axes.

Step 3: For $x > -3$ (the right arm, rising from the vertex at $(-3, 0)$ up to height 3 on the y-axis, i.e. passing through $(0, 3)$): the slope is $\dfrac{3-0}{0-(-3)} = \dfrac{3}{3} = 1$.

Step 4: At the vertex itself, $x = -3$, the two arms meet at a sharp corner (perpendicular, as marked) – so just like the modulus-function corners in Type 15, the derivative does not exist exactly at that single point.

Final Answer

$f'(x) = -1$ for $x < -3$; $f'(x) = 1$ for $x > -3$; $f'(-3)$ does not exist

Common Mistake: Trying to assign a single derivative value that works for the whole graph – a piecewise-linear graph with a corner ALWAYS has a different constant slope on each side, and no derivative at all exactly at the corner.

Quick Recall: On any straight segment of a graph, the derivative is just that segment's slope $\frac{\text{rise}}{\text{run}}$, read directly off the picture – and at any sharp corner where two different slopes meet, the derivative simply doesn't exist there.

IV. Differentiation of Inverse Trigonometric Functions

Type 25

Stating the domain over which each of the six inverse trigonometric functions is differentiable – a purely conceptual/recall type, since some inverse trig functions fail to be differentiable at the very edges of their domain (where the derivative formula would blow up to infinity).

Source: Exercise 5.6, Q1

Given: Find the domain of differentiability of: (i) $\sin^{-1} x$ (ii) $\cos^{-1} x$ (iii) $\sec^{-1} x$ (iv) $\csc^{-1} x$ (v) t

Solution:

Case: (i) $\sin^{-1} x$

Derivative is $\dfrac{1}{\sqrt{1-x^2}}$, which blows up (denominator $\to 0$) exactly at $x = \pm 1$ – so differentiable on $(-1, 1)$, even though the function itself is defined (and continuous) on the closed $[-1, 1]$.

Case: (ii) $\cos^{-1} x$

Same denominator $\sqrt{1-x^2}$ appears – differentiable on $(-1, 1)$.

Case: (iii) $\sec^{-1} x$

Derivative is $\dfrac{1}{|x|\sqrt{x^2-1}}$, undefined at $x = \pm 1$ (denominator zero) and the function itself isn't even defined on $(-1, 1)$ – differentiable on $(-\infty, -1) \cup (1, \infty)$.

Case: (iv) $\csc^{-1} x$

Same reasoning as $\sec^{-1} x$ – differentiable on $(-\infty, -1) \cup (1, \infty)$.

Case: (v) $\tan^{-1} x$

Derivative is $\dfrac{1}{1+x^2}$, which is never zero in the denominator for any real x – differentiable on all of $\mathbf{R}$.

Case: (vi) $\cot^{-1} x$

Same denominator form as $\tan^{-1} x$ – differentiable on all of $\mathbf{R}$.

Final Answer

(i) $(-1, 1)$ (ii) $(-1, 1)$ (iii) $(-\infty, -1) \cup (1, \infty)$ (iv) $(-\infty, -1) \cup (1, \infty)$ (v) $\mathbf{R}$ (vi) $\mathbf{R}$

Common Mistake: Assuming the domain of DIFFERENTIABILITY is the same as the domain of the function itself – for $\sin^{-1} x$ and $\cos^{-1} x$ specifically, the function is defined (and continuous) on the closed interval $[-1, 1]$, but NOT differentiable at the two endpoints ± 1.

Quick Recall: $\sin^{-1}$ and $\cos^{-1}$ lose differentiability at ± 1 (where $\sqrt{1 - x^2} = 0$); $\sec^{-1}$ and $\csc^{-1}$ are simply undefined between -1 and 1; $\tan^{-1}$ and $\cot^{-1}$ are the "safe" ones, differentiable absolutely everywhere.

Type 26

Directly applying the chain rule to differentiate a composition involving an inverse trig function, with no simplification needed first – just the standard inverse-trig derivative formula plus the chain rule.

★ **Board Favorite** – Direct chain-rule application to inverse trig functions is tested constantly as a baseline skill before the more elaborate substitution-based questions.

Formula Used: $\dfrac{d}{dx} \sin^{-1} u = \dfrac{1}{\sqrt{1 - u^2}} \cdot \dfrac{du}{dx}$, $\dfrac{d}{dx} \tan^{-1} u = \dfrac{1}{1 + u^2} \cdot \dfrac{du}{dx}$

Source: Exercise 5.6, Q2(iv)

Given: $y = \sin(\tan^{-1} x)$

Solution:

Step 1: Outer function is $\sin(\cdot)$, inner function is $\tan^{-1} x$ – apply the chain rule.

Step 2: $\dfrac{dy}{dx} = \cos(\tan^{-1} x) \cdot \dfrac{d}{dx}(\tan^{-1} x) = \cos(\tan^{-1} x) \cdot \dfrac{1}{1 + x^2}$.

Step 3: To simplify $\cos(\tan^{-1} x)$: picture a right triangle where the angle $\theta = \tan^{-1} x$ has opposite side x and adjacent side 1, giving hypotenuse $\sqrt{1 + x^2}$; so $\cos\theta = \dfrac{1}{\sqrt{1 + x^2}}$.

Step 4: Substitute back: $\dfrac{dy}{dx} = \dfrac{1}{\sqrt{1 + x^2}} \cdot \dfrac{1}{1 + x^2} = \dfrac{1}{(1 + x^2)^{3/2}}$.

Final Answer

$$\frac{dy}{dx} = \frac{1}{(1 + x^2)^{3/2}}$$

Common Mistake: Leaving the answer as $\cos(\tan^{-1} x) \cdot \dfrac{1}{1+x^2}$ without simplifying $\cos(\tan^{-1} x)$ into an algebraic expression in x using the triangle method – most board mark schemes expect the fully simplified algebraic form.

Alternate Board-Style Example

Source: Exercise 5.6, Q10(i)

Given: $y = x\tan^{-1} x$

Solution:

Step 1: This is a PRODUCT of x and $\tan^{-1} x$ – apply the product rule: $\dfrac{d}{dx}(uv) = u'v + uv'$.

Step 2: Let $u = x$ (so $u' = 1$) and $v = \tan^{-1} x$ (so $v' = \dfrac{1}{1+x^2}$, a direct chain-rule/standard-derivative application).

Step 3: $\dfrac{dy}{dx} = (1)(\tan^{-1} x) + x\left(\dfrac{1}{1+x^2}\right) = \tan^{-1} x + \dfrac{x}{1+x^2}$.

Final Answer

$$\frac{dy}{dx} = \tan^{-1} x + \frac{x}{1+x^2}$$

Quick Recall: Apply the ordinary product/chain rules exactly as with any other function, using the six standard inverse-trig derivative formulas as building blocks – and simplify any trig(inverse-trig(x)) combination using the right-triangle method.

Type 27

Simplifying an inverse trig expression using a KNOWN inverse-trig identity (like $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$, or $\cot^{-1}(\tan x) = \frac{\pi}{2} - x$) BEFORE differentiating – often collapsing the whole expression to something trivial.

★ **Board Favorite** – Recognising a disguised standard identity before differentiating (rather than grinding through the chain rule) is a key time-saving skill that boards specifically reward.

Formula Used: $\sin^{-1} x + \cos^{-1} x = \dfrac{\pi}{2}$ for all $x \in [-1, 1]$

Source: Exercise 5.6, Q5

Given: $y = \sin^{-1} x + \cos^{-1} x$. Find $\dfrac{dy}{dx}$.

Solution:

Step 1: Recognise the standard identity: $\sin^{-1} x + \cos^{-1} x = \dfrac{\pi}{2}$ for every x in the common domain $[-1, 1]$ – a CONSTANT, not a genuine function of x.

Step 2: Since y is just the constant $\dfrac{\pi}{2}$ in disguise, its derivative is immediately 0 – no chain rule computation is even needed.

Final Answer

$$\frac{dy}{dx} = 0$$

Common Mistake: Differentiating $\sin^{-1} x$ and $\cos^{-1} x$ separately (getting $\frac{1}{\sqrt{1-x^2}}$ and $\frac{-1}{\sqrt{1-x^2}}$) without noticing that adding them was ALWAYS going to give zero – while not technically wrong, this wastes significant time versus spotting the identity first.

Alternate Board-Style Example

Source: Exercise 5.6, Q4

Given: $f(x) = \cot^{-1}(\tan x)$. Show that $f'(x) = -1$.

Solution:

Step 1: Use the identity $\cot^{-1}(\tan x) = \dfrac{\pi}{2} - x$ (valid on an appropriate restricted domain) – this comes from the general complementary-angle relationship $\cot^{-1}\theta = \dfrac{\pi}{2} - \tan^{-1}\theta$ applied with $\theta = \tan x$, together with $\tan^{-1}(\tan x) = x$.

Step 2: So $f(x) = \dfrac{\pi}{2} - x$, a simple linear function.

Step 3: Differentiate directly: $f'(x) = 0 - 1 = -1$.

Final Answer

$$f'(x) = -1, \text{ as required}$$

Quick Recall: Before applying the chain rule to any inverse-trig expression, scan for a standard identity (complementary pairs summing to $\frac{\pi}{2}$, or an inverse function directly undoing its own trig function) that might collapse the whole expression to something linear or constant.

Type 28

Simplifying an inverse trig expression using a TRIGONOMETRIC SUBSTITUTION (commonly $x = \tan\theta$, $x = \sin\theta$, or $x = \cos 2\theta$) before differentiating – this is the single most heavily-tested technique in the entire inverse-trig differentiation topic.

★ **Board Favorite** – Trigonometric-substitution simplification is by far the most frequently repeated pattern across ISC and NCERT inverse-trig questions – nearly every "hard" inverse-trig derivative question in the book is solved this way.

Formula Used: Substitute $x = \tan\theta$ (when $1 + x^2$ appears), or $x = \sin\theta$ (when $1 - x^2$ appears)

Source: Exercise 5.6, Q16(i)

Given: $y = \tan^{-1}\left(\dfrac{\sqrt{1+x^2}-1}{x}\right)$

Solution:

Step 1: The presence of $\sqrt{1+x^2}$ signals the substitution $x = \tan\theta$, so that $\sqrt{1+x^2} = \sqrt{1+\tan^2\theta} = \sec\theta$ – a standard Pythagorean simplification.

Step 2: Substitute: $y = \tan^{-1}\left(\dfrac{\sec\theta - 1}{\tan\theta}\right)$.

Step 3: Rewrite $\sec\theta$ and $\tan\theta$ in terms of sine and cosine: $\dfrac{\sec\theta-1}{\tan\theta} = \dfrac{\frac{1}{\cos\theta}-1}{\frac{\sin\theta}{\cos\theta}} = \dfrac{1-\cos\theta}{\sin\theta}$.

Step 4: Use the half-angle identities $1-\cos\theta = 2\sin^2\dfrac{\theta}{2}$ and $\sin\theta = 2\sin\dfrac{\theta}{2}\cos\dfrac{\theta}{2}$: $\dfrac{2\sin^2(\theta/2)}{2\sin(\theta/2)\cos(\theta/2)} = \tan\dfrac{\theta}{2}$.

Step 5: So $y = \tan^{-1}\left(\tan\dfrac{\theta}{2}\right) = \dfrac{\theta}{2}$, and since $\theta = \tan^{-1} x$, we get $y = \dfrac{1}{2}\tan^{-1} x$.

Step 6: Differentiate this much simpler form: $\dfrac{dy}{dx} = \dfrac{1}{2}\cdot\dfrac{1}{1+x^2}$.

Final Answer

$$\frac{dy}{dx} = \frac{1}{2(1+x^2)}$$

Common Mistake: Attempting to differentiate the original messy expression directly with the chain and quotient rules together – this leads to extremely messy algebra; recognising the substitution first turns a multi-page computation into a two-line one.

Alternate Board-Style Example

Source: Exercise 5.6, Q18(ii) (NCERT)

Given: $y = \sin^{-1}\left(\dfrac{1-x^2}{1+x^2}\right)$

Solution:

Step 1: The combination $\dfrac{1-x^2}{1+x^2}$ strongly signals $x = \tan\theta$, since $\dfrac{1-\tan^2\theta}{1+\tan^2\theta} = \cos 2\theta$ (the standard double-angle identity for cosine in terms of tangent).

Step 2: Substitute: $y = \sin^{-1}(\cos 2\theta)$.

Step 3: Rewrite $\cos 2\theta = \sin\left(\dfrac{\pi}{2} - 2\theta\right)$ (co-function identity), so $y = \sin^{-1}\left(\sin\left(\dfrac{\pi}{2} - 2\theta\right)\right) = \dfrac{\pi}{2} - 2\theta$ (valid within the principal range).

Step 4: Since $\theta = \tan^{-1} x$, we get $y = \dfrac{\pi}{2} - 2\tan^{-1} x$.

Step 5: Differentiate: $\dfrac{dy}{dx} = 0 - 2 \cdot \dfrac{1}{1+x^2} = \dfrac{-2}{1+x^2}$.

Final Answer

$$\frac{dy}{dx} = \frac{-2}{1+x^2}$$

Quick Recall: Spot the Pythagorean/double-angle SHAPE hiding in the expression ($1+x^2 \to$ use $x = \tan\theta$; $1-x^2 \to$ use $x = \sin\theta$), substitute, collapse using a standard trig identity, un-substitute back to $\tan^{-1} x$, and only THEN differentiate the now-simple expression. This same substitution instinct also applies to triangle-ratio problems (e.g. simplifying $\sec^{-1}\frac{1}{\sqrt{1-x^2}}$ by picturing a right triangle) – always simplify before differentiating.

V. Implicit Differentiation

Type 29

Differentiating an equation relating x and y directly (implicitly), where y cannot easily be isolated as an explicit function of x – apply the chain rule to every term containing y, remembering that $\frac{d}{dx}(y^n) = ny^{n-1}\frac{dy}{dx}$, then solve algebraically for $\frac{dy}{dx}$.

★ **Board Favorite** – Implicit differentiation is a core, constantly-tested skill – essential whenever y appears mixed with x in a way that can't be cleanly separated.

Formula Used: $\dfrac{d}{dx}(y^n) = ny^{n-1}\dfrac{dy}{dx}$ (chain rule applied to any term containing y)

Source: Exercise 5.7, Q8(v)

Given: $xy + y^2 = \tan x + y$. Find $\dfrac{dy}{dx}$.

Solution:

Step 1: Differentiate both sides term by term with respect to x. For xy (a product of two functions of x), use the product rule: $\dfrac{d}{dx}(xy) = 1 \cdot y + x \cdot \dfrac{dy}{dx} = y + x\dfrac{dy}{dx}$.

Step 2: For y^2, apply the chain rule: $\dfrac{d}{dx}(y^2) = 2y\dfrac{dy}{dx}$.

Step 3: The right side: $\dfrac{d}{dx}(\tan x) = \sec^2 x$ and $\dfrac{d}{dx}(y) = \dfrac{dy}{dx}$.

Step 4: Assembling the full differentiated equation: $y + x\dfrac{dy}{dx} + 2y\dfrac{dy}{dx} = \sec^2 x + \dfrac{dy}{dx}$.

Step 5: Collect all $\dfrac{dy}{dx}$ terms on one side: $x\dfrac{dy}{dx} + 2y\dfrac{dy}{dx} - \dfrac{dy}{dx} = \sec^2 x - y \Rightarrow \dfrac{dy}{dx}(x+2y-1) = \sec^2 x - y$.

Step 6: Divide to isolate $\dfrac{dy}{dx}$: $\dfrac{dy}{dx} = \dfrac{\sec^2 x - y}{x + 2y - 1}$.

Final Answer

$$\frac{dy}{dx} = \frac{\sec^2 x - y}{x + 2y - 1}$$

Common Mistake: Forgetting to apply the chain rule's extra factor $\dfrac{dy}{dx}$ to the lone "y" term on the right-hand side – EVERY occurrence of y anywhere in the equation, no matter how simple, contributes a $\frac{dy}{dx}$ when differentiated.

Alternate Board-Style Example

Source: Exercise 5.7, Q8(vi)

Given: $y = \tan(x + y)$. Find $\dfrac{dy}{dx}$.

Solution:

Step 1: Differentiate both sides. The right side needs the chain rule since the argument $(x + y)$ itself depends on y (which depends on x): $\dfrac{dy}{dx} = \sec^2(x + y) \cdot \dfrac{d}{dx}(x + y) = \sec^2(x + y)\left(1 + \dfrac{dy}{dx}\right)$.

Step 2: Expand the right side: $\dfrac{dy}{dx} = \sec^2(x + y) + \sec^2(x + y)\dfrac{dy}{dx}$.

Step 3: Collect the $\dfrac{dy}{dx}$ terms: $\dfrac{dy}{dx} - \sec^2(x + y)\dfrac{dy}{dx} = \sec^2(x + y) \Rightarrow \dfrac{dy}{dx}\left[1 - \sec^2(x + y)\right] = \sec^2(x + y)$.

Step 4: Divide: $\dfrac{dy}{dx} = \dfrac{\sec^2(x + y)}{1 - \sec^2(x + y)}$.

Step 5: Simplify the denominator using $1 - \sec^2\theta = -\tan^2\theta$: $\dfrac{dy}{dx} = \dfrac{\sec^2(x + y)}{-\tan^2(x + y)} = -\dfrac{1}{\sin^2(x + y)}$ (using $\frac{\sec^2\theta}{\tan^2\theta} = \frac{1}{\sin^2\theta}$) $= -\csc^2(x + y)$.

Final Answer

$$\frac{dy}{dx} = -\csc^2(x + y)$$

Quick Recall: Treat y as "some unknown function of x" throughout – every time you differentiate a term containing y, the chain rule tacks on a $\frac{dy}{dx}$ factor; collect all such terms on one side at the end and divide to isolate $\frac{dy}{dx}$.

Type 30

Verifying the reciprocal relationship $\frac{dy}{dx} \cdot \frac{dx}{dy} = 1$ for an implicitly-defined relation – differentiate once treating y as a function of x (for dy/dx), and once treating x as a function of y (for dx/dy), then multiply.

Source: Exercise 5.7, Q7(i)

Given: $y^2 = 4ax$. Verify that $\dfrac{dy}{dx} \cdot \dfrac{dx}{dy} = 1$.

Solution:

Step 1: **Find $\dfrac{dy}{dx}$:** differentiate $y^2 = 4ax$ with respect to x, treating y as a function of x: $2y\dfrac{dy}{dx} = 4a \Rightarrow \dfrac{dy}{dx} = \dfrac{4a}{2y} = \dfrac{2a}{y}$.

Step 2: **Find $\dfrac{dx}{dy}$:** now differentiate the SAME equation $y^2 = 4ax$ with respect to y instead, treating x as a function of y: $2y = 4a\dfrac{dx}{dy} \Rightarrow \dfrac{dx}{dy} = \dfrac{2y}{4a} = \dfrac{y}{2a}$.

Step 3: Multiply the two results together: $\dfrac{dy}{dx} \cdot \dfrac{dx}{dy} = \dfrac{2a}{y} \cdot \dfrac{y}{2a} = 1$.

Final Answer

$\dfrac{dy}{dx} \cdot \dfrac{dx}{dy} = 1$, verified

Common Mistake: Trying to find $\dfrac{dx}{dy}$ by simply flipping the already-computed $\dfrac{dy}{dx}$ upside down algebraically instead of actually re-differentiating the original equation with respect to y – while the reciprocal relationship IS always true, the exercise specifically wants both derivatives computed independently to demonstrate it.

Quick Recall: To find $\frac{dx}{dy}$, differentiate the ORIGINAL equation with respect to y this time (swapping which variable is "independent") rather than algebraically inverting an already-known $\frac{dy}{dx}$ – then multiply the two results to confirm they're reciprocals.

Type 31

Differentiating an infinite nested radical or continued expression (like $y = \sqrt{x + \sqrt{x + \sqrt{x + \cdots \infty}}}$) – the key trick is recognising that the SAME infinite expression reappears inside itself, letting you replace the inner infinite tail with y itself, turning it into an ordinary implicit equation.

★ **Board Favorite –** The "infinite nesting equals itself" substitution trick is a distinctive, frequently-tested higher-order pattern in this chapter.

Source: Exercise 5.7, Q12(i)

Given: $y = \sqrt{x + \sqrt{x + \sqrt{x + \cdots \infty}}}$. Prove that $(2y-1)\dfrac{dy}{dx} = 1$.

Solution:

Step 1: The expression under the very first square root is $x + \sqrt{x + \sqrt{x + \cdots \infty}}$ – but this inner infinite tail is EXACTLY the same infinite expression as y itself (since removing one layer from an infinite nest leaves the same infinite nest).

Step 2: So we can rewrite the whole thing as simply $y = \sqrt{x+y}$ – collapsing the infinite expression into a simple implicit equation in x and y.

Step 3: Square both sides to remove the outer root: $y^2 = x + y$.

Step 4: Differentiate implicitly with respect to x: $2y\dfrac{dy}{dx} = 1 + \dfrac{dy}{dx}$.

Step 5: Collect the $\dfrac{dy}{dx}$ terms: $2y\dfrac{dy}{dx} - \dfrac{dy}{dx} = 1 \Rightarrow (2y-1)\dfrac{dy}{dx} = 1$.

Final Answer

$(2y-1)\dfrac{dy}{dx} = 1$, as required

Common Mistake: Trying to differentiate the infinite expression "layer by layer" directly – this is impossible since there are infinitely many layers; the entire trick is to notice the self-similar structure FIRST and collapse it into a finite equation before differentiating at all.

Alternate Board-Style Example

Source: Exercise 5.7, Q12(ii)

Given: $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \cdots \infty}}}$. Prove that $(2y-1)\dfrac{dy}{dx} = \cos x$.

Solution:

Step 1: Using the same self-similarity trick, the inner infinite tail equals y itself, so $y = \sqrt{\sin x + y}$.

Step 2: Square both sides: $y^2 = \sin x + y$.

Step 3: Differentiate implicitly with respect to x: $2y\dfrac{dy}{dx} = \cos x + \dfrac{dy}{dx}$.

Step 4: Collect terms: $2y\dfrac{dy}{dx} - \dfrac{dy}{dx} = \cos x \Rightarrow (2y-1)\dfrac{dy}{dx} = \cos x$.

Final Answer

$(2y-1)\dfrac{dy}{dx} = \cos x, \text{ as required}$

Quick Recall: For any infinitely-nested self-similar expression, replace the entire inner infinite tail with y itself (since removing one layer from infinity leaves infinity unchanged), turning an impossible-looking limit into an ordinary implicit equation you can differentiate normally.

VI. Differentiation of Exponential and Logarithmic Functions

Type 32

Directly applying the chain rule to differentiate a composite exponential or logarithmic function – using $\frac{d}{dx}e^u = e^u \cdot u'$ or $\frac{d}{dx}\log u = \frac{u'}{u}$ as the base building block.

★ **Board Favorite –** Chain-rule differentiation of exponential/log compositions is a core, constantly recurring baseline skill in this section, tagged directly to NCERT.

Formula Used: $\dfrac{d}{dx}e^u = e^u \cdot \dfrac{du}{dx}, \quad \dfrac{d}{dx}\log u = \dfrac{1}{u} \cdot \dfrac{du}{dx}$

Source: Exercise 5.8, Q4(i) (NCERT)

Given: $y = e^{x^3}$

Solution:

Step 1: Outer function is $e^{(\cdot)}$, inner function is x^3.

Step 2: $\dfrac{dy}{dx} = e^{x^3} \cdot \dfrac{d}{dx}(x^3) = e^{x^3} \cdot 3x^2$.

Final Answer

$\dfrac{dy}{dx} = 3x^2 e^{x^3}$

Common Mistake: Forgetting to multiply by the derivative of the inner function ($3x^2$) and just writing e^{x^3} as the final answer – the chain rule's extra factor is easy to drop when the inner function is itself a simple power.

Alternate Board-Style Example

Source: Exercise 5.8, Q6(ii)

Given: $y = \tan^{-1}(e^{\sin x})$

Solution:

Step 1: Three layers here: $\tan^{-1}(\cdot)$, then $e^{(\cdot)}$, then $\sin x$ – work from the outside in.

Step 2: Differentiate the outer $\tan^{-1}$ layer: $\dfrac{dy}{dx} = \dfrac{1}{1+(e^{\sin x})^2} \cdot \dfrac{d}{dx}(e^{\sin x}) = \dfrac{1}{1+e^{2\sin x}} \cdot \dfrac{d}{dx}(e^{\sin x})$.

Step 3: Differentiate the middle exponential layer: $\dfrac{d}{dx}(e^{\sin x}) = e^{\sin x} \cdot \cos x$.

Step 4: Combine: $\dfrac{dy}{dx} = \dfrac{e^{\sin x}\cos x}{1+e^{2\sin x}}$.

Final Answer

$\dfrac{dy}{dx} = \dfrac{e^{\sin x}\cos x}{1+e^{2\sin x}}$

Quick Recall: Work from the outside layer inward exactly as with any chain-rule problem – the exponential/log rules are just two more "outer function" types to add to the same peeling-back process.

Type 33

Checking whether a logarithmic (or similarly domain-restricted) function is derivable at a specific point – sometimes the point in question isn't even IN the function's domain at all, which immediately settles the question without any derivative computation.

Source: Exercise 5.8, Q7

Given: Is the function $f(x) = \log(2x-1)$ derivable at $x = 0$?

Solution:

Step 1: Before attempting any derivative computation, check whether $x = 0$ is even in the domain of f – the domain requires the log's argument to be strictly positive: $2x - 1 > 0 \Rightarrow x > \dfrac{1}{2}$.

Step 2: Since $0 \not> \dfrac{1}{2}$, the point $x = 0$ does NOT belong to the domain of f at all – $f(0)$ is not even defined.

Step 3: A function cannot possibly be differentiable at a point where it isn't even defined.

Final Answer

No, f is not derivable at $x = 0$, since $x = 0$ does not even lie in the domain of f

Common Mistake: Jumping straight into differentiating $\log(2x-1)$ (getting $\frac{2}{2x-1}$) and evaluating at $x=0$ without first checking domain membership – this would produce a numeric-looking (but meaningless) answer instead of recognising the question is a domain trick.

Quick Recall: Before differentiating ANY function at a specific point, first check that the point actually lies in the function's domain – this is especially critical for log and root functions, where the "obvious" computation can be run even where it shouldn't be.

Type 34

Differentiating a composite expression involving an ABSTRACT function f (with no explicit formula), using only given numeric values like $f(a)$ and $f'(a)$ – apply the chain rule symbolically, then substitute the given numbers at the very end.

★ **Board Favorite** – Differentiating abstract functions from given data points tests genuine understanding of the chain rule (as opposed to rote formula-plugging), making it a recurring higher-order favorite.

Source: Exercise 5.8, Q10

Given: $f(1)=4,\ f'(1)=2$. Find the derivative of $\log(f(e^x))$ w.r.t. x at $x=0$.

Solution:

Step 1: Let $y=\log(f(e^x))$. Differentiate using the chain rule, treating f as an unknown but differentiable function: $\dfrac{dy}{dx}=\dfrac{1}{f(e^x)}\cdot f'(e^x)\cdot e^x$ – three layers: log, then f, then e^x, each contributing its own factor.

Step 2: We need this at $x=0$: substitute $x=0$ throughout, noting $e^0=1$: $\left.\dfrac{dy}{dx}\right|_{x=0}=\dfrac{1}{f(1)}\cdot f'(1)\cdot e^0=\dfrac{f'(1)}{f(1)}\cdot 1$.

Step 3: Substitute the given numeric values: $\dfrac{f'(1)}{f(1)}=\dfrac{2}{4}=\dfrac{1}{2}$.

Final Answer

$\dfrac{1}{2}$

Common Mistake: Trying to substitute $x = 0$ before finishing the symbolic differentiation – always differentiate the ENTIRE expression symbolically first (keeping f and f' as abstract symbols), and only plug in the specific point and given numeric values as the very last step.

Alternate Board-Style Example

Source: Exercise 5.8, Q11

Given: $f(x) = e^x g(x),\ g(0) = 2,\ g'(0) = 1$. Find $f'(0)$.

Solution:

Step 1: This is a PRODUCT of e^x and $g(x)$ – apply the product rule symbolically: $f'(x) = e^x \cdot g(x) + e^x \cdot g'(x) = e^x(g(x) + g'(x))$.

Step 2: Substitute $x = 0$ (noting $e^0 = 1$): $f'(0) = 1 \cdot (g(0) + g'(0))$.

Step 3: Substitute the given values: $f'(0) = 2 + 1 = 3$.

Final Answer

$f'(0) = 3$

Quick Recall: Differentiate the whole expression symbolically FIRST, keeping the abstract function's name and its derivative as placeholders throughout – substitute the specific point and given numeric values only in the very last step.

Type 35

Simplifying an exponential expression using logarithm rules (like $e^{\log u} = u$, or $a \log b = \log b^a$) BEFORE differentiating, rather than differentiating the messy original expression directly.

Source: Exercise 5.8, Q18(ii)

Given: $y = e^{2\log x + 3x}$. Prove that $\dfrac{dy}{dx} = x(2 + 3x)e^{3x}$.

Solution:

Step 1: Split the exponent using $e^{A+B} = e^A \cdot e^B$: $y = e^{2\log x} \cdot e^{3x}$.

Step 2: Simplify $e^{2\log x}$ using $a \log b = \log b^a$ first: $2\log x = \log x^2$, so $e^{2\log x} = e^{\log x^2} = x^2$ (since $e^{\log u} = u$ for $u > 0$).

Step 3: So y simplifies dramatically: $y = x^2 e^{3x}$ – an ordinary product, far simpler than the original exponential form.

Step 4: Differentiate using the product rule: $\dfrac{dy}{dx} = 2x \cdot e^{3x} + x^2 \cdot 3e^{3x} = e^{3x}(2x + 3x^2)$.

Step 5: Factor out x: $= xe^{3x}(2 + 3x)$.

Final Answer

$\dfrac{dy}{dx} = x(2 + 3x)e^{3x}$, as required

Common Mistake: Trying to differentiate $e^{2\log x + 3x}$ directly via the chain rule without first simplifying $e^{2\log x}$ back down to x^2 – this leads to an unnecessarily complicated (though not technically wrong) derivative that's much harder to match to the target answer.

Quick Recall: Whenever you see $e^{(\dots + k\log x + \dots)}$, split the exponent and use $e^{k\log x} = x^k$ to simplify BEFORE differentiating – this often converts a scary-looking exponential into an ordinary polynomial-times-exponential product.

Type 36

Implicit differentiation of an equation that mixes exponential and logarithmic terms of both x and y together – apply implicit differentiation as usual, but expect log/exponential chain-rule factors on top of the usual $\frac{dy}{dx}$ bookkeeping.

★ **Board Favorite** – Mixed exponential-log implicit differentiation is a recurring higher-order pattern, with the book tagging a fuller version as a Competency Focused Question.

Source: Exercise 5.8, Q20(ii)

Given: $e^{x-y} = \log\left(\dfrac{x}{y}\right)$. Find $\dfrac{dy}{dx}$.

Solution:

Step 1: Simplify the right side first using $\log(x/y) = \log x - \log y$ – easier to differentiate as a difference than as a single quotient-log.

Step 2: Differentiate the left side: $\dfrac{d}{dx}e^{x-y} = e^{x-y}\left(1 - \dfrac{dy}{dx}\right)$ – chain rule, since the exponent $x - y$ depends on y too.

Step 3: Differentiate the right side: $\dfrac{d}{dx}(\log x - \log y) = \dfrac{1}{x} - \dfrac{1}{y}\dfrac{dy}{dx}$.

Step 4: Equate: $e^{x-y}\left(1 - \dfrac{dy}{dx}\right) = \dfrac{1}{x} - \dfrac{1}{y}\dfrac{dy}{dx}$.

Step 5: Expand and collect $\frac{dy}{dx}$ terms on one side: $e^{x-y} - e^{x-y}\frac{dy}{dx} = \frac{1}{x} - \frac{1}{y}\frac{dy}{dx} \Rightarrow \frac{1}{y}\frac{dy}{dx} - e^{x-y}\frac{dy}{dx} = \frac{1}{x} - e^{x-y}$.

Step 6: Factor and solve: $\frac{dy}{dx}\left(\frac{1}{y} - e^{x-y}\right) = \frac{1}{x} - e^{x-y} \Rightarrow \frac{dy}{dx} = \frac{\frac{1}{x} - e^{x-y}}{\frac{1}{y} - e^{x-y}}$, which simplifies (multiplying numerator and denominator by xy) to $\frac{y - xye^{x-y}}{x - xye^{x-y}} = \frac{y(1 - xe^{x-y})}{x(1 - ye^{x-y})}$.

Final Answer

$$\frac{dy}{dx} = \frac{y(1 - xe^{x-y})}{x(1 - ye^{x-y})}$$

Common Mistake: Trying to differentiate $\log(x/y)$ as a single quotient-log using the quotient rule for logs directly instead of first splitting it into $\log x - \log y$ – the split form is always easier to differentiate term by term.

Alternate Board-Style Example

Source: Exercise 5.8, Q21 (Competency Focused Question)

Given: $\log(x^2 + y^2) = 2\tan^{-1}\frac{y}{x}$. Show that $\frac{dy}{dx} = \frac{x+y}{x-y}$.

Solution:

Step 1: Differentiate the left side: $\frac{d}{dx}\log(x^2 + y^2) = \frac{1}{x^2 + y^2}\left(2x + 2y\frac{dy}{dx}\right)$.

Step 2: Differentiate the right side using the standard $\tan^{-1}$ derivative with $u = y/x$: $\frac{d}{dx}\left(2\tan^{-1}\frac{y}{x}\right) = 2 \cdot \frac{1}{1 + (y/x)^2} \cdot \frac{d}{dx}\left(\frac{y}{x}\right)$. First, $\frac{1}{1 + (y/x)^2} = \frac{x^2}{x^2 + y^2}$. Second, by the quotient rule, $\frac{d}{dx}\left(\frac{y}{x}\right) = \frac{x\frac{dy}{dx} - y}{x^2}$.

Step 3: Combine the right side: $2 \cdot \frac{x^2}{x^2 + y^2} \cdot \frac{x\frac{dy}{dx} - y}{x^2} = \frac{2\left(x\frac{dy}{dx} - y\right)}{x^2 + y^2}$.

Step 4: Equate left and right sides (both already have denominator $x^2 + y^2$, so it cancels immediately): $2x + 2y\frac{dy}{dx} = 2x\frac{dy}{dx} - 2y$.

Step 5: Collect $\frac{dy}{dx}$ terms: $2y\frac{dy}{dx} - 2x\frac{dy}{dx} = -2y - 2x \Rightarrow \frac{dy}{dx}(2y - 2x) = -2(x + y)$.

Step 6: Divide: $\frac{dy}{dx} = \frac{-2(x+y)}{2(y-x)} = \frac{-(x+y)}{y-x} = \frac{x+y}{x-y}$.

Final Answer

$\dfrac{dy}{dx} = \dfrac{x+y}{x-y}$, as required

Quick Recall: Split logs of products/quotients into sums/differences before differentiating, and for $\tan^{-1}(y/x)$ terms remember the denominator $1 + (y/x)^2$ conveniently simplifies to $\frac{x^2+y^2}{x^2}$ – often cancelling nicely against a matching $x^2 + y^2$ on the other side.

Type 37

Differentiating a logarithm with a variable (non-e) BASE, such as $\log_x(\cdot)$ or $\log_{g(x)}(\cdot)$ – convert to natural logarithms using the change-of-base formula FIRST, since there is no direct "variable-base log" derivative rule.

★ **Board Favorite –** Variable-base logarithm differentiation is a distinctive recurring pattern that specifically tests whether a student remembers the change-of-base conversion step.

Formula Used: $\log_a b = \dfrac{\log b}{\log a}$ (change of base, converting to natural/common logarithms)

Source: Exercise 5.8, Q23(i)

Given: $y = \log_x(2x-3)$. Differentiate w.r.t. x.

Solution:

Step 1: There is no standard derivative formula for a log with a VARIABLE base like x – first convert using change of base: $y = \log_x(2x-3) = \dfrac{\log(2x-3)}{\log x}$ – now it's an ordinary quotient of two natural-log functions of x.

Step 2: Apply the quotient rule: $\dfrac{dy}{dx} = \dfrac{\log x \cdot \frac{d}{dx}\log(2x-3) - \log(2x-3)\cdot\frac{d}{dx}\log x}{(\log x)^2}$.

Step 3: Compute the two derivatives needed: $\dfrac{d}{dx}\log(2x-3) = \dfrac{2}{2x-3}$ (chain rule) and $\dfrac{d}{dx}\log x = \dfrac{1}{x}$.

Step 4: Substitute: $\dfrac{dy}{dx} = \dfrac{\log x \cdot \frac{2}{2x-3} - \log(2x-3)\cdot\frac{1}{x}}{(\log x)^2}$.

Step 5: Combine over a common denominator inside the numerator (multiplying through by $x(2x-3)$ in numerator and denominator together) to reach a cleaner single-fraction form: $\dfrac{dy}{dx} = \dfrac{2x\log x - (2x-3)\log(2x-3)}{x(2x-3)(\log x)^2}$.

Final Answer

$$\frac{dy}{dx} = \frac{2x\log x - (2x-3)\log(2x-3)}{x(2x-3)(\log x)^2}$$

Common Mistake: Trying to differentiate $\log_x(2x-3)$ directly as if x were a fixed constant base (treating it like $\log_2(2x-3)$) – since the base ITSELF is a function of x here, change of base to convert into an ordinary quotient of natural logs is essential first.

Alternate Board-Style Example

Source: Exercise 5.8, Q23(ii)

Given: $y = \log_{\cos x}(\sin x)$. Differentiate w.r.t. x.

Solution:

Step 1: Change of base first: $y = \dfrac{\log(\sin x)}{\log(\cos x)}$.

Step 2: Quotient rule: $\dfrac{dy}{dx} = \dfrac{\log(\cos x)\cdot \frac{d}{dx}\log(\sin x) - \log(\sin x)\cdot \frac{d}{dx}\log(\cos x)}{(\log\cos x)^2}$.

Step 3: Compute the pieces: $\dfrac{d}{dx}\log(\sin x) = \dfrac{\cos x}{\sin x} = \cot x$ and $\dfrac{d}{dx}\log(\cos x) = \dfrac{-\sin x}{\cos x} = -\tan x$.

Step 4: Substitute: $\dfrac{dy}{dx} = \dfrac{\log(\cos x)\cdot\cot x - \log(\sin x)\cdot(-\tan x)}{(\log\cos x)^2} = \dfrac{\cot x\log(\cos x) + \tan x\log(\sin x)}{(\log\cos x)^2}$.

Final Answer

$$\frac{dy}{dx} = \frac{\cot x\log\cos x + \tan x\log\sin x}{(\log\cos x)^2}$$

Quick Recall: Any log with a non-constant (variable) base MUST be converted via change-of-base into a quotient of two natural logs before any differentiation begins – there is no shortcut derivative formula for a variable-base logarithm.

VII. Logarithmic Differentiation / Power-Tower Functions

Type 38

Differentiating a product of several factors (polynomial or trigonometric) by taking the logarithm of both sides first – this converts a long product into a SUM of logs, which is far easier to differentiate term by term than applying the product rule repeatedly.

★ **Board Favorite** – Logarithmic differentiation of multi-factor products is a core, frequently-tested technique, tagged directly to NCERT in the book.

Formula Used: For $y = f_1(x)f_2(x)\cdots f_n(x)$: take log of both sides $\Rightarrow \log y = \sum \log f_i(x),$ then differentiate implicitly

Source: Exercise 5.9, Q1(i) (NCERT)

Given: $y = (x+3)^2(x+4)^3(x+5)^4$

Solution:

Step 1: Take log of both sides: $\log y = 2\log(x+3) + 3\log(x+4) + 4\log(x+5)$ – using $\log(AB) = \log A + \log B$ and $\log(A^n) = n\log A$ repeatedly.

Step 2: Differentiate both sides implicitly with respect to x. The left side needs the chain rule: $\frac{1}{y}\frac{dy}{dx}$.

Step 3: The right side differentiates term by term: $\frac{2}{x+3} + \frac{3}{x+4} + \frac{4}{x+5}$.

Step 4: So $\frac{1}{y}\frac{dy}{dx} = \frac{2}{x+3} + \frac{3}{x+4} + \frac{4}{x+5}$.

Step 5: Multiply both sides by y (substituting back the original expression for y): $\frac{dy}{dx} = (x+3)^2(x+4)^3(x+5)^4\left[\frac{2}{x+3} + \frac{3}{x+4} + \frac{4}{x+5}\right]$.

Final Answer

$$\frac{dy}{dx} = (x+3)^2(x+4)^3(x+5)^4\left[\frac{2}{x+3} + \frac{3}{x+4} + \frac{4}{x+5}\right]$$

Common Mistake: Trying to apply the ordinary product rule directly to three (or more) factors – this is technically possible but extremely error-prone with this many terms; taking logs first converts the product into a much more manageable sum.

Alternate Board-Style Example

Source: Exercise 5.9, Q1(ii) (NCERT)

Given: $y = \cos x \cos 2x \cos 3x$

Solution:

Step 1: Take log of both sides: $\log y = \log \cos x + \log \cos 2x + \log \cos 3x$.

Step 2: Differentiate implicitly: $\frac{1}{y}\frac{dy}{dx} = \frac{-\sin x}{\cos x} + \frac{-2\sin 2x}{\cos 2x} + \frac{-3\sin 3x}{\cos 3x} = -\tan x - 2\tan 2x - 3\tan 3x$.

Step 3: Multiply both sides by y: $\frac{dy}{dx} = -\cos x \cos 2x \cos 3x(\tan x + 2\tan 2x + 3\tan 3x)$.

Final Answer

$$\frac{dy}{dx} = -\cos x \cos 2x \cos 3x(\tan x + 2\tan 2x + 3\tan 3x)$$

Quick Recall: Whenever a function is a PRODUCT of several factors (especially three or more, or with awkward powers), take logs of both sides first to turn the product into a sum – differentiate that sum term by term, then multiply back by the original y.

Type 39

The master "power-tower" type: differentiating a function of the form $[f(x)]^{g(x)}$, where the variable appears in BOTH the base and the exponent – neither the power rule nor the exponential rule alone applies, so logarithmic differentiation is essential.

★ **Board Favorite** – This is one of the single most heavily-tested patterns in the entire differentiation syllabus – variable-base-variable-exponent functions appear in nearly every board paper.

Formula Used: For $y = [f(x)]^{g(x)}$: $\log y = g(x)\log f(x)$, then differentiate implicitly (product rule needed on the right)

Source: Exercise 5.9, Q3(ii) (NCERT)

Given: $y = (\sin x)^{\sin x}, \ 0 < x < \pi$

Solution:

Step 1: Take log of both sides: $\log y = \sin x \cdot \log(\sin x)$ – using $\log(A^B) = B \log A$.

Step 2: Differentiate implicitly. The left side: $\frac{1}{y}\frac{dy}{dx}$. The right side is now a PRODUCT of $\sin x$ and $\log(\sin x)$, so apply the product rule: $\frac{d}{dx}[\sin x \log(\sin x)] = \cos x \cdot \log(\sin x) + \sin x \cdot \frac{\cos x}{\sin x}$.

Step 3: Simplify the second term: $\sin x \cdot \frac{\cos x}{\sin x} = \cos x$.

Step 4: So $\frac{1}{y}\frac{dy}{dx} = \cos x \log(\sin x) + \cos x = \cos x[\log(\sin x) + 1]$.

Step 5: Multiply both sides by $y = (\sin x)^{\sin x}$: $\frac{dy}{dx} = (\sin x)^{\sin x} \cos x[1 + \log(\sin x)]$.

Final Answer

$$\frac{dy}{dx} = (\sin x)^{\sin x} \cos x \left[1 + \log(\sin x)\right]$$

Common Mistake: Trying to use the ordinary power rule (nx^{n-1}, treating the exponent as constant) or the ordinary exponential rule ($a^x \log a$, treating the base as constant) – since BOTH the base and exponent contain x here, neither rule alone applies; only logarithmic differentiation works.

Alternate Board-Style Example

Source: Exercise 5.9, Q4(ii)

Given: $y = (2x+3)^{x-5},\ x > -\frac{3}{2}$

Solution:

Step 1: Take log: $\log y = (x-5)\log(2x+3)$.

Step 2: Differentiate implicitly, using the product rule on the right (with $u = x-5$, $v = \log(2x+3)$): $\frac{1}{y}\frac{dy}{dx} = 1 \cdot \log(2x+3) + (x-5) \cdot \frac{2}{2x+3}$.

Step 3: Multiply both sides by $y = (2x+3)^{x-5}$: $\frac{dy}{dx} = (2x+3)^{x-5}\left[\log(2x+3) + \frac{2(x-5)}{2x+3}\right]$.

Final Answer

$$\frac{dy}{dx} = (2x+3)^{x-5}\left[\log(2x+3) + \frac{2(x-5)}{2x+3}\right]$$

Quick Recall: Whenever you see (variable)$^{(\text{variable})}$ – the variable appearing in BOTH the base and the exponent – take logs of both sides immediately, apply the product rule to the resulting $g(x)\log f(x)$, then multiply back by the original function.

Type 40

A single combined question testing whether a student can correctly distinguish and differentiate all FOUR distinct forms in one sum: x^x (both variable), a^x (constant base), x^a (constant exponent), and a^a (a genuine constant) – each needs a completely different technique.

★ **Board Favorite** – This "four forms in one sum" combination question is a classic, frequently-repeated pattern that specifically tests whether a student applies the RIGHT rule to each term rather than one blanket approach.

Source: Exercise 5.9, Q7 (NCERT)

Given: Find the derivative of $x^x + a^x + x^a + a^a$ for some fixed constant $a > 0, x > 0$.

Solution:

Step 1: **Term** x^x (both base and exponent are the variable x): needs logarithmic differentiation. Let $u = x^x$; $\log u = x\log x$; differentiate: $\dfrac{1}{u}\dfrac{du}{dx} = \log x + x\cdot\dfrac{1}{x} = \log x + 1$; so $\dfrac{du}{dx} = x^x(1+\log x)$.

Step 2: **Term** a^x (constant base a, variable exponent x): use the standard exponential rule directly, $\dfrac{d}{dx}a^x = a^x\log a$.

Step 3: **Term** x^a (variable base x, constant exponent a): use the ordinary POWER rule (since the exponent is a fixed constant here, not a variable), $\dfrac{d}{dx}x^a = ax^{a-1}$.

Step 4: **Term** a^a (both base and exponent are fixed constants, so this whole term is just a CONSTANT number): its derivative is 0.

Step 5: Add all four derivatives together.

Final Answer

$$\frac{d}{dx}\left(x^x + a^x + x^a + a^a\right) = x^x(1+\log x) + a^x\log a + ax^{a-1}$$

Common Mistake: Treating all four terms with the SAME rule (most commonly, mistakenly applying logarithmic differentiation to every term, including a^x and x^a where a simpler direct rule already applies, or worse, treating a^a as a variable term and trying to "differentiate" it instead of recognising it's a plain constant).

Quick Recall: Before differentiating each term in a mixed sum, ask separately for each one: "is the variable in the base, the exponent, both, or neither?" – that answer alone tells you whether to use logarithmic differentiation, the exponential rule, the power rule, or simply zero (constant).

Type 41

An IMPLICIT equation where power-tower terms appear on both sides (e.g. $x^y + y^x = a^b$, or two different bases raised to swapped variable exponents) – take logs of the appropriate terms, then differentiate implicitly as usual, being extra careful with the product rule on each power-tower term.

★ **Board Favorite** – Implicit power-tower equations are one of the most demanding (and most-tested) combinations in this chapter, requiring logarithmic AND implicit differentiation together.

Source: Exercise 5.9, Q10(ii)

Given: $x^y + y^x = a^b$ (a fixed constant). Find $\frac{dy}{dx}$.

Solution:

Step 1: Since the right side a^b is a genuine constant (both a, b fixed), its derivative is simply 0 – this significantly simplifies the final equation.

Step 2: Handle x^y (variable base, variable exponent) via logarithmic differentiation: let $u = x^y$, so $\log u = y \log x$. Differentiate: $\frac{1}{u}\frac{du}{dx} = \frac{dy}{dx}\log x + y \cdot \frac{1}{x}$, so $\frac{du}{dx} = x^y\left(\frac{dy}{dx}\log x + \frac{y}{x}\right)$.

Step 3: Handle y^x similarly: let $v = y^x$, so $\log v = x \log y$. Differentiate: $\frac{1}{v}\frac{dv}{dx} = \log y + x \cdot \frac{1}{y}\frac{dy}{dx}$, so $\frac{dv}{dx} = y^x\left(\log y + \frac{x}{y}\frac{dy}{dx}\right)$.

Step 4: Add the two derivatives and set equal to 0: $x^y\left(\frac{dy}{dx}\log x + \frac{y}{x}\right) + y^x\left(\log y + \frac{x}{y}\frac{dy}{dx}\right) = 0$.

Step 5: Collect all $\frac{dy}{dx}$ terms on one side: $\frac{dy}{dx}\left(x^y \log x + \frac{xy^x}{y}\right) = -\left(\frac{yx^y}{x} + y^x \log y\right)$.

Step 6: Solve for $\frac{dy}{dx}$ and simplify by multiplying numerator and denominator by xy to clear the small fractions: $\frac{dy}{dx} = -\frac{y(yx^y + xy^x \log y)}{x(x^y \log x \cdot y + xy^x)}$, which is more cleanly written as $\frac{dy}{dx} = -\frac{y^x \log y + yx^{y-1}}{x^y \log x + xy^{x-1}}$ after re-grouping the original derivative terms directly.

Final Answer

$$\frac{dy}{dx} = -\frac{y^x \log y + yx^{y-1}}{x^y \log x + xy^{x-1}}$$

Common Mistake: Applying the SAME logarithmic-differentiation setup to both x^y and y^x without being careful about which variable is the "base" and which is the "exponent" in each term – a small mix-up here (swapping x and y's roles) silently produces the wrong formula.

Alternate Board-Style Example

Source: Exercise 5.9, Q10(vi)

Given: $(\cos x)^y = (\cos y)^x$. Find $\dfrac{dy}{dx}$.

Solution:

Step 1: Take log of both sides directly (since it's already an equation, not a sum): $y \log(\cos x) = x \log(\cos y)$.

Step 2: Differentiate both sides implicitly, using the product rule on each side. Left side: $\dfrac{dy}{dx} \log(\cos x) + y \cdot \dfrac{-\sin x}{\cos x} = \dfrac{dy}{dx} \log(\cos x) - y \tan x$.

Step 3: Right side: $1 \cdot \log(\cos y) + x \cdot \dfrac{-\sin y}{\cos y} \dfrac{dy}{dx} = \log(\cos y) - x \tan y \dfrac{dy}{dx}$.

Step 4: Equate: $\dfrac{dy}{dx} \log(\cos x) - y \tan x = \log(\cos y) - x \tan y \dfrac{dy}{dx}$.

Step 5: Collect $\dfrac{dy}{dx}$ terms: $\dfrac{dy}{dx} [\log(\cos x) + x \tan y] = \log(\cos y) + y \tan x$.

Step 6: Divide: $\dfrac{dy}{dx} = \dfrac{\log(\cos y) + y \tan x}{\log(\cos x) + x \tan y}$.

Final Answer

$$\frac{dy}{dx} = \frac{\log(\cos y) + y \tan x}{\log(\cos x) + x \tan y}$$

Quick Recall: For implicit equations with power towers on both sides, take logs of the WHOLE equation (not term by term, if it's already an equality of two power-tower expressions) and carefully apply the product rule, tracking which variable is being differentiated (contributing $\frac{dy}{dx}$) and which is the plain independent variable x.

Type 42

An infinite, self-referential power tower (like $y = x^{x^{x^{\cdots\infty}}}$) – exactly as with infinite nested radicals (Type 31), the key trick is recognising that the infinite exponent tower reappears inside itself, so the WHOLE expression can be replaced by y itself, collapsing it to $y = x^y$.

Source: Exercise 5.9, Q11

Given: $y = x^{x^{x^{\cdots\infty}}}$. Prove that $\dfrac{dy}{dx} = \dfrac{y^2}{x(1 - y\log x)}$.

Solution:

Step 1: The exponent of the very first x is $x^{x^{x^{\cdots\infty}}}$ itself – exactly the same infinite tower as y. So the whole expression collapses to $y = x^y$.

Step 2: Take log of both sides: $\log y = y \log x$.

Step 3: Differentiate implicitly, using the product rule on the right (since both y and $\log x$ depend on x): $\dfrac{1}{y}\dfrac{dy}{dx} = \dfrac{dy}{dx}\log x + y \cdot \dfrac{1}{x}$.

Step 4: Collect the $\dfrac{dy}{dx}$ terms on one side: $\dfrac{1}{y}\dfrac{dy}{dx} - \dfrac{dy}{dx}\log x = \dfrac{y}{x} \Rightarrow \dfrac{dy}{dx}\left(\dfrac{1}{y} - \log x\right) = \dfrac{y}{x}$.

Step 5: Multiply through by y to clear the fraction inside the bracket: $\dfrac{dy}{dx}(1 - y\log x) = \dfrac{y^2}{x}$.

Step 6: Divide: $\dfrac{dy}{dx} = \dfrac{y^2}{x(1 - y\log x)}$.

Final Answer

$\dfrac{dy}{dx} = \dfrac{y^2}{x(1 - y\log x)}$, as required

Common Mistake: Trying to differentiate the infinite tower "one exponent at a time" – exactly as with infinite radicals, this is impossible with infinitely many layers; the self-similarity substitution ($y = x^y$) must be spotted FIRST.

Quick Recall: An infinite power tower always equals itself one layer down – replace the whole infinite exponent with y to get a simple implicit equation ($y = x^y$), take logs, then differentiate implicitly as normal.

VIII. Differentiation of Parametric Functions

Type 43

Standard parametric differentiation – when both x and y are given as separate functions of a third parameter t (or θ), find $\frac{dy}{dx}$ by differentiating each with respect to the parameter separately, then dividing.

★ **Board Favorite –** Standard parametric differentiation is the foundational skill of this section, tagged directly to NCERT, and underlies every other parametric question type.

Formula Used: $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}, \ \dfrac{dx}{dt} \neq 0$

Source: Exercise 5.10, Q1(ii) (NCERT)

Given: $x = 2at^2, \ y = at^4$

Solution:

Step 1: Differentiate x with respect to t: $\dfrac{dx}{dt} = 4at$.

Step 2: Differentiate y with respect to t: $\dfrac{dy}{dt} = 4at^3$.

Step 3: Divide: $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt} = \dfrac{4at^3}{4at} = t^2$.

Final Answer

$\dfrac{dy}{dx} = t^2$

Common Mistake: Attempting to eliminate the parameter t first and find y as an explicit function of x before differentiating – this is unnecessary extra work (and sometimes impossible); parametric differentiation lets you find $\frac{dy}{dx}$ directly, still in terms of t, without ever eliminating the parameter.

Alternate Board-Style Example

Source: Exercise 5.10, Q7(i)

Given: $x = e^t(\sin t + \cos t),\ y = e^t(\sin t - \cos t)$

Solution:

Step 1: Differentiate x using the product rule: $\dfrac{dx}{dt} = e^t(\sin t + \cos t) + e^t(\cos t - \sin t) = e^t[(\sin t + \cos t) + (\cos t - \sin t)] = 2e^t \cos t$ – the $\sin t$ terms cancel.

Step 2: Differentiate y similarly: $\dfrac{dy}{dt} = e^t(\sin t - \cos t) + e^t(\cos t + \sin t) = e^t[(\sin t - \cos t) + (\cos t + \sin t)] = 2e^t \sin t$ – the $\cos t$ terms cancel this time.

Step 3: Divide: $\dfrac{dy}{dx} = \dfrac{2e^t \sin t}{2e^t \cos t} = \tan t$.

Final Answer

$\dfrac{dy}{dx} = \tan t$

Quick Recall: Never try to eliminate the parameter – differentiate x and y each separately with respect to t (product rule if needed), then simply divide $\frac{dy/dt}{dx/dt}$; parameters usually cancel out beautifully in the final ratio.

Type 44

Differentiating one given function of x with respect to ANOTHER given function of x (not with respect to plain x) – treat both as "parametric" functions of the underlying variable x acting as the hidden parameter, differentiate each separately, then divide.

★ **Board Favorite** – "Differentiate A with respect to B" (rather than with respect to x) is a distinctive, frequently-tested question format that specifically tests whether a student recognises the parametric-style approach.

Formula Used: To differentiate $u(x)$ w.r.t. $v(x)$: compute $\dfrac{du}{dx}$ and $\dfrac{dv}{dx}$ separately, then $\dfrac{du}{dv} = \dfrac{du/dx}{dv/dx}$

Source: Exercise 5.10, Q12(ii)

Given: Differentiate $\sin x^2$ w.r.t. x^3

Solution:

Step 1: Let $u = \sin(x^2)$ and $v = x^3$ – we want $\dfrac{du}{dv}$, NOT $\dfrac{du}{dx}$ directly.

Step 2: Differentiate u with respect to x: $\dfrac{du}{dx} = \cos(x^2) \cdot 2x$ (chain rule).

Step 3: Differentiate v with respect to x: $\dfrac{dv}{dx} = 3x^2$.

Step 4: Divide: $\dfrac{du}{dv} = \dfrac{du/dx}{dv/dx} = \dfrac{2x\cos(x^2)}{3x^2} = \dfrac{2\cos(x^2)}{3x}$.

Final Answer

$$\frac{2\cos(x^2)}{3x}$$

Common Mistake: Misreading "differentiate A w.r.t. B" as an ordinary "differentiate A w.r.t. x" question and simply computing $\frac{du}{dx}$ alone – the actual target is the RATIO $\frac{du/dx}{dv/dx}$, treating x as the hidden connecting parameter.

Alternate Board-Style Example

Source: Exercise 5.10, Q12(iv)

Given: Differentiate $\sin^2 x$ w.r.t. $e^{\cos x}$

Solution:

Step 1: Let $u = \sin^2 x$ and $v = e^{\cos x}$.

Step 2: $\dfrac{du}{dx} = 2\sin x\cos x$ (chain rule on $\sin^2 x$).

Step 3: $\dfrac{dv}{dx} = e^{\cos x} \cdot (-\sin x)$ (chain rule on the exponential).

Step 4: Divide: $\dfrac{du}{dv} = \dfrac{2\sin x\cos x}{-\sin x \cdot e^{\cos x}} = \dfrac{-2\cos x}{e^{\cos x}}$ – the $\sin x$ factor cancels.

Final Answer

$$\frac{-2\cos x}{e^{\cos x}}$$

Quick Recall: "Differentiate A with respect to B" (when neither is the plain variable x) always means: find $\frac{dA}{dx}$ and $\frac{dB}{dx}$ separately (using x as the hidden connecting parameter), then divide – exactly like ordinary parametric differentiation.

Type 45

Differentiating one inverse-trigonometric expression with respect to ANOTHER inverse-trigonometric expression – both expressions typically need their own trigonometric substitution and simplification FIRST (as in Section IV), and only after both are reduced to simple linear forms in the substituted angle should the ratio be taken.

★ **Board Favorite** – This combines two of the chapter's hardest skills (trig-substitution simplification AND parametric-style differentiation) into one question, making it a classic high-value board long-answer type.

Source: Exercise 5.10, Q16(ii)

Given: Differentiate $\tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$ w.r.t. $\cos^{-1}(2x\sqrt{1-x^2})$

Solution:

Step 1: Let $u = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$ and $v = \cos^{-1}(2x\sqrt{1-x^2})$. Substitute $x = \sin\theta$ in BOTH (a natural choice since $\sqrt{1-x^2} = \cos\theta$ appears in both expressions).

Step 2: Simplify u: $u = \tan^{-1}\left(\frac{\cos\theta}{\sin\theta}\right) = \tan^{-1}(\cot\theta) = \tan^{-1}\left(\tan\left(\frac{\pi}{2}-\theta\right)\right) = \frac{\pi}{2}-\theta$.

Step 3: Simplify v: $2x\sqrt{1-x^2} = 2\sin\theta\cos\theta = \sin 2\theta$, so $v = \cos^{-1}(\sin 2\theta) = \cos^{-1}\left(\cos\left(\frac{\pi}{2}-2\theta\right)\right) = \frac{\pi}{2}-2\theta$.

Step 4: Both u and v are now simple linear functions of the SAME parameter θ (with $\theta = \sin^{-1}x$ as the hidden connector) – differentiate each with respect to θ: $\frac{du}{d\theta} = -1$ and $\frac{dv}{d\theta} = -2$.

Step 5: Divide: $\frac{du}{dv} = \frac{du/d\theta}{dv/d\theta} = \frac{-1}{-2} = \frac{1}{2}$.

Final Answer

$\frac{1}{2}$

Common Mistake: Trying to differentiate the original messy inverse-trig expressions directly with respect to x using the chain and quotient rules – this produces extremely messy algebra; substituting $x = \sin\theta$ in BOTH expressions first (using the same θ for both) reduces the whole problem to comparing two simple linear functions of θ.

Quick Recall: When asked to differentiate one inverse-trig expression with respect to another, try the SAME substitution in both (usually $x = \sin\theta$ or $x = \tan\theta$, whichever matches the recurring radical/fraction pattern) – if both collapse to linear functions of the substituted angle, the final ratio is just the ratio of their slopes.

IX. Second Order Derivatives

Type 46

Finding the second derivative of an explicit function directly – differentiate once to get $\frac{dy}{dx}$, then differentiate that result again with respect to x to get $\frac{d^2y}{dx^2}$.

★ **Board Favorite** – Direct second-derivative computation is the foundational skill of this section, tagged directly to NCERT.

Formula Used: $\dfrac{d^2y}{dx^2} = \dfrac{d}{dx}\left(\dfrac{dy}{dx}\right)$

Source: Exercise 5.11, Q1(iii) (NCERT)

Given: $y = \tan^{-1} x$

Solution:

Step 1: First derivative: $\dfrac{dy}{dx} = \dfrac{1}{1+x^2}$ – the standard inverse-trig derivative.

Step 2: Rewrite as $(1+x^2)^{-1}$ to differentiate again using the chain rule: $\dfrac{d^2y}{dx^2} = -1 \cdot (1+x^2)^{-2} \cdot 2x = \dfrac{-2x}{(1+x^2)^2}$.

Final Answer

$$\frac{d^2y}{dx^2} = \frac{-2x}{(1+x^2)^2}$$

Common Mistake: Forgetting to apply the chain rule (the extra factor $2x$ from differentiating the inner $1+x^2$) when differentiating $(1+x^2)^{-1}$ a second time – treating it like a simple power of x alone instead of a composite function.

Alternate Board-Style Example

Source: Exercise 5.11, Q4(iv) (NCERT)

Given: $y = \sin^3 x$

Solution:

Step 1: First derivative: $\dfrac{dy}{dx} = 3\sin^2 x \cos x$ (chain rule, treating $\sin^3 x = (\sin x)^3$).

Step 2: Differentiate again using the product rule (on $3\sin^2 x$ and $\cos x$): $\dfrac{d^2y}{dx^2} = 3\left[2\sin x \cos x \cdot \cos x + \sin^2\right.$
$3\left[2\sin x \cos^2 x - \sin^3 x\right]$.

Step 3: Simplify: $= 6\sin x \cos^2 x - 3\sin^3 x$.

Final Answer

$\dfrac{d^2y}{dx^2} = 6\sin x \cos^2 x - 3\sin^3 x$

Quick Recall: Compute $\frac{dy}{dx}$ fully and simplify it first – then differentiate that (often product-rule-shaped) expression a second time, watching for BOTH the product rule and any remaining chain rule factors.

Type 47

Proving a stated second-order differential-equation-style identity (typically of the form $ay''+by'+cy=0$, or with variable coefficients like $x^2y''+xy'+y=0$) – differentiate the given function twice, substitute both derivatives into the stated identity, and confirm it simplifies to the claimed result.

★ **Board Favorite** – This is the single biggest board-favorite cluster in the whole chapter – "differentiate twice and prove the DE" questions (especially involving $\log x$, $\cos(\log x)$, or $e^{a\cos^{-1}x}$ style functions) appear in nearly every ISC paper, with the book tagging a direct instance to ISC 2024.

Source: Exercise 5.11, Q12(i)

Given: $y = \sin(\log x)$. Prove that $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0$.

Solution:

Step 1: First derivative (chain rule): $\dfrac{dy}{dx} = \cos(\log x)\cdot\dfrac{1}{x} = \dfrac{\cos(\log x)}{x}$.

Step 2: Before differentiating again, multiply both sides by x to simplify the upcoming product-rule step: $x\dfrac{dy}{dx} = \cos(\log x)$.

Step 3: Differentiate THIS equation with respect to x instead (avoiding the quotient rule): left side needs the product rule: $\dfrac{d}{dx}\left(x\dfrac{dy}{dx}\right) = \dfrac{dy}{dx}+x\dfrac{d^2y}{dx^2}$. Right side: $\dfrac{d}{dx}\cos(\log x) = -\sin(\log x)\cdot\dfrac{1}{x} = \dfrac{-\sin(\log x)}{x} = \dfrac{-y}{x}$ (since $y = \sin(\log x)$ by definition).

Step 4: So $\dfrac{dy}{dx} + x\dfrac{d^2y}{dx^2} = \dfrac{-y}{x}$.

Step 5: Multiply both sides by x to clear the remaining fraction: $x\dfrac{dy}{dx} + x^2\dfrac{d^2y}{dx^2} = -y$.

Step 6: Rearrange: $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0$.

Final Answer

$x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0$, as required

Common Mistake: Differentiating the ORIGINAL first derivative $\frac{\cos(\log x)}{x}$ directly using the quotient rule – this works but produces messier algebra; multiplying through by x FIRST (to get $x\frac{dy}{dx} = \cos(\log x)$) and then using the product rule on the left is significantly cleaner and is the standard technique for this entire family of questions.

Alternate Board-Style Example

Source: Exercise 5.11, Q12(ii) (ISC 2024)

Given: $y = 3\cos(\log x) + 4\sin(\log x)$. Prove that $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0$.

Solution:

Step 1: First derivative: $\dfrac{dy}{dx} = -3\sin(\log x)\cdot\dfrac{1}{x} + 4\cos(\log x)\cdot\dfrac{1}{x} = \dfrac{-3\sin(\log x) + 4\cos(\log x)}{x}$.

Step 2: Multiply by x first: $x\dfrac{dy}{dx} = -3\sin(\log x) + 4\cos(\log x)$.

Step 3: Differentiate again (product rule on the left, direct chain rule on the right): $\dfrac{dy}{dx} + x\dfrac{d^2y}{dx^2} = -3\cos(\log x)\cdot\dfrac{1}{x} - 4\sin(\log x)\cdot\dfrac{1}{x} = \dfrac{-3\cos(\log x) - 4\sin(\log x)}{x} = \dfrac{-y}{x}$ (recognising the original y hiding in the numerator, with a sign flip).

Step 4: Multiply through by x: $x\dfrac{dy}{dx} + x^2\dfrac{d^2y}{dx^2} = -y$.

Step 5: Rearrange: $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0$.

Final Answer

$x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0$, as required

Quick Recall: For any y built from $\sin(\log x)/\cos(\log x)$ combinations, multiply the first derivative by x BEFORE differentiating again – this avoids the quotient rule and almost always lets the original y reappear directly (with a sign) inside the second-derivative equation, ready to rearrange into the required identity.

Type 48

Finding the second derivative of a parametric function – since $\frac{dy}{dx}$ itself comes out as a function of the parameter t (not of x directly), it must be differentiated AGAIN with respect to t first, and only then divided by $\frac{dx}{dt}$ – never differentiate $\frac{dy}{dx}$ directly with respect to x.

★ **Board Favorite** – Second-order parametric differentiation is a distinctive, frequently-tested combination skill, with the book tagging a direct instance to the ISC Specimen Paper 2024.

Formula Used: $\dfrac{d^2y}{dx^2} = \dfrac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt}$

Source: Exercise 5.11, Q24 (ISC Specimen Paper 2024)

Given: $x = a\sin(pt),\ y = b\cos(pt)$. Find the value of $\dfrac{d^2y}{dx^2}$ at $t = 0$.

Solution:

Step 1: First derivatives w.r.t. t: $\dfrac{dx}{dt} = ap\cos(pt)$ and $\dfrac{dy}{dt} = -bp\sin(pt)$.

Step 2: Form the first parametric derivative: $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt} = \dfrac{-bp\sin(pt)}{ap\cos(pt)} = \dfrac{-b}{a}\tan(pt)$ – the p cancels.

Step 3: To get the second derivative, differentiate THIS result with respect to t (NOT with respect to x directly): $\dfrac{d}{dt}\left(\dfrac{dy}{dx}\right) = \dfrac{-b}{a} \cdot p\sec^2(pt)$.

Step 4: Divide by $\dfrac{dx}{dt}$ again to convert back to a derivative with respect to x: $\dfrac{d^2y}{dx^2} = \dfrac{-\frac{bp}{a}\sec^2(pt)}{ap\cos(pt)} = \dfrac{-b\sec^2(pt)}{a^2\cos(pt)} = \dfrac{-b\sec^3(pt)}{a^2}$ (since $\frac{\sec^2\theta}{\cos\theta} = \sec^3\theta$).

Step 5: Evaluate at $t = 0$: $\sec(p\cdot 0) = \sec 0 = 1$, so $\sec^3(0) = 1$.

Final Answer

$$\left.\frac{d^2y}{dx^2}\right|_{t=0} = \frac{-b}{a^2}$$

Common Mistake: Differentiating $\frac{dy}{dx}$ directly with respect to x as if it were an ordinary function of x – since $\frac{dy}{dx}$ is expressed in terms of the PARAMETER t, it must first be differentiated with respect to t, and only that result divided by $\frac{dx}{dt}$ again.

Alternate Board-Style Example

Source: Exercise 5.11, Q21

Given: $x = a(\theta - \sin\theta),\ y = a(1 + \cos\theta)$. Find $\dfrac{d^2y}{dx^2}$.

Solution:

Step 1: First derivatives w.r.t. θ: $\dfrac{dx}{d\theta} = a(1 - \cos\theta)$ and $\dfrac{dy}{d\theta} = -a\sin\theta$.

Step 2: First parametric derivative: $\dfrac{dy}{dx} = \dfrac{-a\sin\theta}{a(1-\cos\theta)} = \dfrac{-\sin\theta}{1-\cos\theta}$. Using the half-angle identities $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$ and $1 - \cos\theta = 2\sin^2\frac{\theta}{2}$: $\dfrac{dy}{dx} = \dfrac{-2\sin(\theta/2)\cos(\theta/2)}{2\sin^2(\theta/2)} = -\cot\dfrac{\theta}{2}$.

Step 3: Differentiate this with respect to θ (NOT x): $\dfrac{d}{d\theta}\left(-\cot\dfrac{\theta}{2}\right) = \dfrac{1}{2}\csc^2\dfrac{\theta}{2}$ (chain rule, since $\frac{d}{d\theta}\cot(\theta/2) = -\csc^2(\theta/2)\cdot\frac{1}{2}$).

Step 4: Divide by $\dfrac{dx}{d\theta} = a(1-\cos\theta) = 2a\sin^2\dfrac{\theta}{2}$ (using the half-angle identity again): $\dfrac{d^2y}{dx^2} = \dfrac{\frac{1}{2}\csc^2(\theta/2)}{2a\sin^2(\theta/2)} = \dfrac{1}{4a\sin^4(\theta/2)}$ (using $\csc^2(\theta/2) = \frac{1}{\sin^2(\theta/2)}$, so dividing by another $\sin^2(\theta/2)$ gives $\sin^4$ in the denominator).

Final Answer

$$\frac{d^2y}{dx^2} = \frac{1}{4a\sin^4(\theta/2)}$$

Quick Recall: For second-order parametric derivatives, NEVER differentiate $\frac{dy}{dx}$ with respect to x directly – always differentiate it with respect to the PARAMETER first, then divide by $\frac{dx}{dt}$ (or $\frac{dx}{d\theta}$) one more time.